European Commission

JRC TECHNICAL REPORT

Marine Strategy Framework Directive Review and analysis of EU Member States' 2018 reports

Descriptor 6: Sea-floor integrity and Descriptor 1: Benthic habitats

Assessment (Art.8), Good Environmental Status (Art. 9) and Targets (Art. 10)

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2021

EUR 30716 EN

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EU Science Hub
https://ec.europa.eu/jrc

JRC125288

EUR 30716 EN

PDF ISBN 978-92-76-38014-6 ISSN 1831-9424 doi:10.2760/355956

Luxembourg: Publications Office of the European Union, 2021

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How to cite this report: Boschetti S. T., Palialexis A., Connor D., *Marine Strategy Framework Directive, Review and analysis of EU Member States' 2018 reports. Descriptor 6: Sea-floor integrity and Descriptor 1: Benthic habitats,* EUR 30716 EN, Publications Office of the European Union, Luxembourg, 2021, ISBN 978-92-79-38014-6, doi:10.2760/355956, JRC125288.

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Acknowledgements

We thank the European Environment Agency for providing the data reported and the Directorate-General for Environment for their constructive comments to improve the impact of the analysis.

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Foreword

The Marine Directors of the European Union and all EU Member States have jointly developed a common strategy for supporting the implementation of the “Marine Strategy Framework Directive” (MSFD), 2008/56/EC, amended by Commission Directive (EU) 2017/845 of 17 May 2017.

The European Commission Joint Research Centre is delivering thematic technical reports to support MSFD implementation, such as guidance documents, technical background reports and analyses related to EU Member States reporting. These thematic reports are targeted at experts who are directly or indirectly implementing the MSFD and support the further development of the Directive.

The JRC's technical report series “Marine Strategy Framework Directive - Review and analysis of EU Member States' 2018 reports”, analyse, from a technical point of view, the MSFD reports submitted by EU Member States pursuant to MSFD Article 17. The analysis includes the GES Assessment (Article 8), the determination of Good Environmental Status (Article 9) and the Targets setting (Article 10). The outcome provides information for the further development of the policy implementation, supported by the EU Member States through established MSFD Technical Groups and Expert Networks.

Abstract

The current report analyses the information provided by the European Union Member States (MS) on Themes – sea-floor integrity and benthic habitats, relating to Descriptors 6 and 1 - of the Marine Strategy Framework Directive (MSFD, EU 2008). The reported information by the MS follows the Article 17 requirements of the MSFD for updating Articles 8, 9 and 10 of the Directive, for the last 6-year MSFD reporting cycle. By September 2020, 20 Member States reported on D6/D1 in electronic format (hereafter e-reports).

1 Introduction

The Descriptor 6 criteria of the Commission Decision (EU) 2017/848 (EU 2017, hereafter GES Decision) address the theme sea-floor integrity and the ecosystem component benthic habitats, relevant for Descriptors 6 and 1 respectively of MSFD Annex I (EU, 2008).

Descriptor 6 states that sea-floor integrity is at a level that ensures that the structure and functions of the ecosystems are safeguarded and benthic ecosystems, in particular, are not adversely affected. The GES Decision requires the assessment of physical loss and disturbance at subdivisional level, reflecting biogeographic differences in species composition of the broad habitat type. The GES Decision Part I includes three primary criteria:

- **D6C1**: Physical loss of the seabed, including intertidal areas (permanent change);
- **D6C2**: Spatial extent and distribution of physical disturbance pressures; and
- **D6C3**: Spatial extent of each habitat type which is adversely affected by physical disturbance, this to be based on the outcomes of D6C2.

The relevant pressures identified are physical loss (due to permanent change of seabed substrate or morphology and to extraction of seabed substrate) and physical disturbance to seabed (temporary or reversible). According to the GES Decision, "Physical loss shall be understood as a permanent change to the seabed which has lasted or is expected to last for a period of two reporting cycles (12 years) or more", while "physical disturbance shall be understood as a change to the seabed from which it can recover if the activity causing the disturbance pressure ceases".

The GES Decision Part II addresses the theme Benthic habitats (relating to Descriptors 1 and 6) and refers back to Part I and criteria D6C1, D6C2 and D6C3, and includes two additional primary criteria:

- **D6C4**: The extent of loss of the habitat type, resulting from anthropogenic pressures, to be based on the outcomes of D6C1.
- **D6C5**: The extent of adverse effects from anthropogenic pressures on the condition of the habitat type, including those from D6C3 and D6C4.

D6C1, D6C2 and D6C3 only assess the extent of the physical loss and disturbance and their impact, while D6C4 and D6C5 address the overall assessment of benthic habitats for Descriptors 1 and 6.

The GES Decision specifies that the assessment for D6C3, D6C4 and D6C5 should be done for the benthic broad habitat types, including their associated biological communities, specified in Table 2 of the GES Decision; these equate to one or more habitat types at level 2 of the European Nature Information System (EUNIS) habitat classification (Evans et al. 2016) .

In the present assessment, the reports of Member States (MS) on D6 for articles 8, 9 and 10 were analysed, looking at:

- convergence of the reported criteria elements and methodological standards with the Commission Decision 2017/848;
- consistency across MS regions and subregions on reported criterion elements;
- gaps and future steps.

The MS data used in this analysis was reported between 15th October 2018 and 1st October 2020, and covers the reports of 20 MS who reported electronically. The three MS which had either not reported or not provided electronic reports by October 2020 (Bulgaria, Greece and the United Kingdom) are not included in the analysis. The reported data for Art. 8, Art. 9 and Art. 10 are included in the files D6-Art.8, D6-Art.9 and D6-Art.10 in Annex I.

Table 1. The names and abbreviations of MSFD regions and subregions, as well as of Members States (MS) allocated to their respective subregion(s). MS highlighted in grey had not submitted an electronic report for Art. 8, 9 and 10 by 1st October 2020 and are not included in the current assessment.

Region	Subregion	Member States [MS]							
Baltic Sea [BAL]	BAL	Denmark [DK]	Estonia [EE]	Finland [FI]	Germany [DE]	Latvia [LV]	Lithuania [LT]	Poland [PL]	Sweden [SE]
North East Atlantic [NEA]	Bay of Biscay and the Iberian Coast [ABI]	France [FR]	Portugal [PT]	Spain [ES]					
	Celtic Seas [ACS]	France [FR]	Ireland [IE]	United Kingdom [UK]					
	Macaronesia [AMA]	Portugal [PT]	Spain [ES]						
	Greater North Sea [ANS]	Belgium [BE]	Denmark [DK]	France [FR]	Germany [DE]	Sweden [SE]	The Netherlands [NL]	United Kingdom [UK]	
Mediterranean Sea [MED]	Adriatic Sea [MAD]	Croatia [HR]	Greece [EL]	Italy [IT]	Slovenia [SI]				
	Aegean-Levantine Sea [MAL]	Cyprus [CY]	Greece [EL]						
	Ionian Sea and Central Mediterranean Sea [MIC]	Greece [EL]	Italy [IT]	Malta [MT]					
	Western Mediterranean Sea [MWE]	France [FR]	Italy [IT]	Spain [ES]	United Kingdom [UK]				
Black Sea [BLK]	BLK	Bulgaria [BG]	Romania [RO]						

2 Results of Art. 8 analysis

2.1 Key results and recommendations from MSFD Article 8 reports

- 20 MS reported for Art. 8 in D6, eight in BAL, nine in NEA, seven in MED and one in BLK. Of these, two MS did not provide information on the assessment methodology.
- Few threshold values were reported despite that some benthic habitats are already assessed under other EU policies or by the Regional Sea Conventions (RSCs). Similarly, very few completed assessments were reported, based on an assessed value, and often covering only some of the GES criteria and MSFD broad habitat types (BHT).
- Few MS used assessments from other EU Directives (Water Framework Directive – WFD, Habitats Directive - HD) for D6C5 (adverse effects).
- Many gaps exist in the assessment methodology, across BHT, in all regions. This includes gaps in data/monitoring, assessment methods and lack of common understanding in the assessment approaches (GES determination, integration rules and assessment methodology for parameters and criteria).
- The reporting and assessment methodology should clearly state when D6 assessments were made under other descriptors (e.g., adverse effects of pressures addressed by other Descriptors), as it not obvious in the current reporting structure.
- The progress made by the MSFD Technical Group on seabed habitats and sea-floor integrity (TG Seabed) could not be reflected in this reporting cycle, as most MS completed their assessments before the establishment of the TG; however, many of the above issues are already being discussed by the TG and will develop recommendations for comparable assessments. These should be included in the update of the Art. 8 assessment guidance.
- A regional or EU agreement on the assessment methods by ecologically relevant areas (subdivision), threshold values, parameters, extensions, integration rules across criteria and parameters, and common assessment approach is greatly needed to harmonise the assessments and provide comparable results. TG Seabed can support the harmonisation of this work.

2.2 Consistency, comparability and adequacy of reported features and elements assessed at EU and regional level

2.2.1 Criterion features and elements

Criteria D6C1 and D6C2 are assessed at feature level and address the pressures physical loss (code 'PresPhyLoss' in e-reports) and physical disturbance (code 'PresPhyDisturbSeabed'); therefore no elements need be reported for these two criteria. Overall, 15 MS (75% of the evaluated reports) reported for D6C1 and D6C2. FI, LT, SE for BAL, HR for MED, and RO for BLK did not report for D6C1 and D6C2. PL addressed both pressures for both criteria, while LV, IT and PL also reported an element.

Several countries reported for D6C1 and D6C2; however, they did not provide any assessment.

Data: Member States 2018 reports (file in Annex: D6-Art.8, columns: feature, elements).

Table 2. Features reported by the evaluated Member States and their geographical association with the MSFD regions and subdivisions for criteria D6C1 and D6C2. MS shaded grey indicates reported features, which were not assessed.

Features reported	BAL	NEA				MED				BLK
	BAL	ANS	ACS	ABI	AMA	MWE	MIC	MAD	MAL	BLK
D6C1 Physical loss of the seabed										
Broad Habitat Types – BHT	LV					IT	IT	IT		
Physical disturbance	PL									
Physical loss	DE DK EE PL	BE DE DK FR NL	FR IE	ES FR	ES PT*	ES FR IT	IT MT	IT SI	CY	
D6C2 Physical disturbance to the seabed										
Broad Habitat Types	LV									
Physical disturbance	DE DK EE PL	BE DE DK FR NL	FR IE	ES FR	ES PT	ES FR IT	IT MT	IT SI	CY	
Physical loss	PL									

*PT reported 'Benthic habitats' as element (code HabBenAll).

The list of the BHT assessed by the MS for D6C3, D6C4 and D6C5 were compared to the list given in Table 2 of the GES Decision. Some MS reported as assessment of 'other benthic habitats' (GES Decision, Table 3), comparable assessments to the BHT. In Table 3 the different colours correspond to the following: Black font: MS that reported BHT which are also listed in the TG Seabed document in Annex II (Circabc[1], SEABED_3-2020-05). Red font: Broad habitat types with distribution in the relevant MS which were not reported. Green font: MS which are not included in the TG Seabed document listing the MSFD broad habitat types across regions, subregions and Member States, but reported these habitats for 2018 Art.8. MS shaded grey when the BHT were reported, but not assessed. Current MS that did not provided e-report (BG and EL) have been included in Table 3 to show which BHT by (sub)region are missing.

Table 3. Benthic broad habitat types reported by the evaluated Member States, and their geographical association with the MSFD regions and subdivisions.

Feature: HabBenBHT D6C3 - D6C4 - D6C5	BAL	NEA				MED				BLK
	BAL	ANS	ACS	ABI	AMA	MWE	MIC	MAD	MAL	BLK
Littoral rock and biogenic reef	PL	DE FR	FR IE	FR PT	PT	FR		SI HR		RO BG
Littoral sediment	PL	DE FR NL	FR IE	FR PT	PT	FR	MT	SI HR		RO BG
Infralittoral rock and biogenic reef	DE DK FI LT PL SE EE LV	BE DK FR SE DE	FR IE	FR PT ES	ES PT	FR ES IT	MT IT EL	HR SI IT EL	EL CY	RO BG

[1] https://circabc.europa.eu/w/browse/85f995b6-819a-4d1b-aecd-630df32f83d6

Feature: HabBenBHT D6C3 - D6C4 - D6C5	BAL	NEA				MED				BLK
	BAL	ANS	ACS	ABI	AMA	MWE	MIC	MAD	MAL	BLK
Infralittoral coarse sediment	DK EE FI LT LV DE	BE DK FR SE DE	FR IE	FR PT	PT	FR		HR SI		RO BG
Infralittoral mixed sediment	DE DK FI LT LV PL EE SE	DK FR SE DE	FR IE	FR PT ES	PT	FR IT	MT IT	HR SI IT	EL CY	RO BG
Infralittoral sand	DE DK FI LT LV PL SE EE	BE DE DK FR SE DE	FR IE	FR PT ES	PT ES	FR ES UK IT	MT IT EL	HR SI IT EL	EL CY	RO BG
Infralittoral mud	DE DK FI LT PL SE EE SE LV	BE DK FR SE DE NL	FR IE	FR PT ES	PT ES	FR ES IT	MT IT EL	HR SI IT	EL CY	RO BG
Circalittoral rock and biogenic reef	DE DK FI LT SE EE LV PL	DK FR SEDE	FR IE	FR PT ES	PT ES	FR ES IT	MT IT	SI IT HR	CY	RO BG
Circalittoral coarse sediment	DK FI LT EE LV PL DE SE	DE DK FR NL SE BE	FR IE	FR PT ES	PT ES	FR ES IT	MT IT	SI HR IT	EL CY	RO BG
Circalittoral mixed sediment	DE DK FI LT LV EE PL SE	DK FR SE DE NL BE	FR IE	FR PT ES	PT	FR IT	MT IT	HR SI IT	EL CY	RO BG
Circalittoral sand	DE DK FI LT LV PL SE EE	DE DK FR NL SE BE	FR IE	FR PT ES	PT ES	FR ES IT	MT IT EL	HR SI IT EL	EL CY	RO BG
Circalittoral mud	DE DK FI LT LV SE EE PL	DE DK FR NL SE BE	FR IE	FR PT ES	PT ES	FR ES IT	MT IT EL	SI EL HR IT	EL	RO BG
Offshore circalittoral rock and biogenic reef	FI SE LV	FR SE DE	FR IE	FR PT ES	PT ES	FR ES IT	IT	IT HR	CY	
Offshore circalittoral coarse sediment	DK FI EE LV PL SE	DE DK FR NL SE BE	FR IE	FR PT ES	PT ES	FR ES IT	IT	IT	EL	BG
Offshore circalittoral mixed sediment	DK FI LT EE LV PL SE	DK FR SE NL BE	FR IE	FR PT ES	PT	IT	MT IT	IT	EL	RO BG
Offshore circalittoral sand	DK FI LT SE EE LV PL	DE DK FR NL SE BE	FR IE	FR PT ES	PT ES	FR ES IT	IT EL	EL HR IT	EL CY	RO BG

Feature: HabBenBHT D6C3 - D6C4 - D6C5	BAL	NEA				MED				BLK
	BAL	ANS	ACS	ABI	AMA	MWE	MIC	MAD	MAL	BLK
Offshore circalittoral mud	DK FI LT SE EE LV PL	DE DK FR NL SE BE	FR IE	FR PT ES	PT ES	FR ES IT	MT IT EL	EL HR IT	EL	RO BG
Upper bathyal rock and biogenic reef		DK SE	FR IE	FR PT ES	PT ES	FR ES IT	MT IT	IT	CY	RO
Upper bathyal sediment		SE DK	FR IE	FR PT ES	PT ES	FR ESIT	MT IT EL	IT HR EL	EL CY	RO BG
Lower bathyal rock and biogenic reef			FR IE	FR PT ES	PT ES	FR ES IT	MT IT	IT	CY	
Lower bathyal sediment			FR IE	FR PT ES	PT ES	FR ES IT	MT IT EL	IT	EL CY	RO BG
Abyssal			FR IE	FR PT ES	PT ES	FR ES IT	MT IT EL		EL	RO BG
Benthic broad habitats*			IE							
Posidonia oceanica meadows**							MT			
None				ES	ES	ES				

(*) IE reported for all the BHT by adding a generic 'Benthic broad habitats' in the element field.
(**) Posidonia is not a BHT but was reported under BHTs by MT.

Table 4. Other benthic habitats reported for D6/D1 and their distribution by MS, region and subregion.

Feature: HabBenOther D6C3 - D6C4 - D6C5	BAL	NEA				MED				BLK
	BAL	ANS	ACS	ABI	AMA	MWE	MIC	MAD	MAL	BLK
[Cystoseira] spp. in eulittoral rockpools										RO
[Zostera] beds in infralittoral sediments	DE									RO
Baltic muddy bottoms of the aphotic zone	DE									
Coastal Zone Sand (NL)		NL								
Coralligenous habitat						IT	IT	IT		
Coastal/Muddy/Sandy habitats (WFD)	DK	DK								
Dogger Bank Sand (NL)		NL								
Estuaries	DK EE	DK								

Feature: HabBenOther D6C3 - D6C4 - D6C5	BAL	NEA				MED				BLK
	BAL	ANS	ACS	ABI	AMA	MWE	MIC	MAD	MAL	BLK
Frrysian Front Shand (NL)		NL								
Large shallow inlets and bays	DK EE	DK								
Maerl beds						IT	IT	IT		
Mudflats and sandflats not covered by seawater at low tide	DK EE	DK								
Offshore Sand (NL)		NL								
Oyster Banks Mud (NL)		NL								
Pontic [*Phyllophora nervosa*] on vertical rock faces in the lower infralittoral										RO
Posidonia beds (*Posidonion oceanicae*)						IT	IT MT*	IT		
Reefs	DE DK EE	DE DK NL			PT					
Sandbanks which are covered by seawater all the time		NL								
Sandbanks which are slightly covered by seawater all the time	DE DK EE	DE DK								
Soft bottom non biogenic habitat						IT	IT	IT		
Sublittoral coarse sediment	DE	DE								
Sublittoral mud		DE								
Submarine structures made by leaking gases		DK								
Submerged or partially submerged sea caves	DK									
White Corals						IT	IT	IT		

*MT assigned this element to the BHT, although it belongs to the 'other habitat type' table and, therefore, listed here.

Findings: Several countries did not report all HabBenBHT which are known to occur in their marine waters (see MS in red font in Table 3). Other MS which reported HabBenBHT (as being present based on both the 2016 EMODnet MSFD broad habitat types map and on feedback from TG Seabed members[2]), they did not assess them (see grey shading in Table 3). A few MS reported HabBenBHT that were not included in the MSFD broad habitat types across regions, subregions and Member States (SEABED_3-2020-05[2] Annex 1, green font in Table 3). MS which

[2] https://circabc.europa.eu/w/browse/85f995b6-819a-4d1b-aecd-630df32f83d6

did not provided e-reports (EL and BG) where included in the table to provide information on the amount of missing data regarding BHT.

Seven MS reported BHTs in BAL, all nine MS in NEA, six MS in MED and RO in BLK. IE and ES reported a generic element that includes all benthic habitat types. IT and EE did not report any BHT, but only other benthic habitat types.

Overall, seven MS reported 25 other benthic habitat types which occur in their marine waters.

Gaps: -

Recommendation for reporting: MT misreported the element '*Posidonia oceanica* meadows' under BHT; this element shares the same code as 'Posidonia beds (*Posidonion oceanicae*)' in the other benthic habitat type list.

Recommendation for implementation: Table 2 in the GES Decision (EU 2017) ensured a high level of consistency in the reported elements. TG Seabed can help further improve the consistency and completeness of the criteria elements (BHTs) reported in the coming reporting/assessment cycles, through a common understanding of which BHTs are known to occur in each MS, region or subregion. According to the GES Decision, all BHTs which occur in an MS's marine waters should be reported, even if they were not assessed.

2.2.2 Element source

The sources of the reported elements are grouped and presented in Table 5. This information gives an overview of how the MS selected the reported elements and is essential to identify possible differences within each region.

Data: Member States 2018 reports (file in Annex: D6-Art.8; column: Element source).

Table 5. Element source by region and MS for BHT (above) and HabBenOther (below) assessed for D6C3, D6C4 and D6C5 sea-floor integrity and benthic habitats.

BHT element source	Element code source	Regions				Total no. of MS
		BAL	NEA	MED	BLK	
EU	BHT	DE, DK, LT, SE	BE, DE, DK, FR, NL, PT, SE	FR, HR, MT, SI	RO	12
	Other		NL	MT		2
HELCOM	BHT	LV				1
OSPAR	BHT		SE			1
	Other		SE			1
National	BHT	FI, LV, PL		CY, HR		5
Other			IE			1
None			ES	ES		1

Other habitat types element source	Element code source	Regions				Total no. of MS
		BAL	NEA	MED	BLK	
EU	EUNIS	DE			RO	2
	HabitatsDirective	EE	DK			2
	HabitatsDirective: habitats	DE, EE	DE, NL			3
	Other	DK	DK	IT		2
OSPAR	Other		NL			1
National	HabitatsDirective	DK				1
Other	EUNIS	DE	DE			2
	HabitatsDirective		PT			1

Findings: Most of the MS used EU as source for the elements assessed for D6C3, D6C4 and D6C5, for both BHT and other habitat types (code source: HD). The code for the element source used for BHT was BHT (MSFD), while NL and MT did not specify this information. LV reported HELCOM as source (for the elements circalittoral sand, circalittoral mud, circalittoral mixed sediments, infralittoral sand, infralittoral mixed sediment and infralittoral coarse sediments), while SE reported OSPAR (for the elements circalittoral mixed sediments, circalittoral coarse sediments, infralittoral mixed sediment, infralittoral coarse sediments, offshore circalittoral mixed sediments, offshore circalittoral coarse sediments and upper bathyal sediment). CY, FI, HR, LV, and PL reported national as element source and HD as element code source.

For the other habitat types for which EU was reported as element source, EUNIS and the HD were assigned as code source.

NL reported OSPAR as source for the elements Coastal Zone Sand, Dogger Bank Sand, Offshore Sand and Oyster Banks Mud.

Gaps: -

Recommendation for reporting: MS should report EU as source when referring to the MSFD BHTs.

Recommendation for implementation: This entry provides an overview of the source of the reported elements. According to the GES Decision, MS should agree (sub)regionally on the 'other habitats' they wish to assess.

2.2.3 Extent of use of MSFD Criteria (primary and secondary)

All D6 criteria are primary. The GES Decision specifies that the assessment of D6C4 should be based on the outcome of the assessment of D6C1, while D6C3 on the outcome of D6C2. The assessment of D6C5 takes account of the outcome of D6C3 and D6C4.

Data: Member States 2018 reports (file in Annex: D6-Art.8; column: Criterion, Parameter achieved).

Table 6. Features and elements reported by MS and criterion per region (the analysis was not conducted by subregion). Assessed element are green highlighted. Note that the reported elements are not always assessed**.** Top: D6C1 and D6C2; Bottom: D6C3, D6C4 and D6C5.

Region	MS	Feature	Element	D6C1	D6C2
BAL	DE	PresPhyLoss	None	X	
		PresPhyDisturbSeabed	None		X
	DK	PresPhyLoss	None	X	
		PresPhyDisturbSeabed	None		X
	EE	PresPhyLoss	None	X	
		PresPhyDisturbSeabed	None		X
	LV	HabBenBHT	None	X	X
	PL	PresPhyLoss	None	X	X
		PresPhyDisturbSeabed	None	X	X
NEA	BE	PresPhyLoss	None	X	
		PresPhyDisturbSeabed	None		X
	DE	PresPhyLoss	None	X	
		PresPhyDisturbSeabed	None		X
	DK	PresPhyLoss	None	X	
		PresPhyDisturbSeabed	None		X
	ES	PresPhyLoss	None	X	
		PresPhyDisturbSeabed	None		X
	FR	PresPhyLoss	None	X	
		PresPhyDisturbSeabed	None		X
	IE	PresPhyLoss	None	X	
		PresPhyDisturbSeabed	None		X
	NL	PresPhyLoss	None	X	
		PresPhyDisturbSeabed	None		X
	PT	PresPhyLoss	Benthic habitats	X	
		PresPhyLoss	None	X	
		PresPhyDisturbSeabed	None		X
MED	CY	PresPhyLoss	None	X	
		PresPhyDisturbSeabed	None		X
	ES	PresPhyLoss	None	X	
		PresPhyDisturbSeabed	None		X
	FR	PresPhyLoss	None	X	
		PresPhyDisturbSeabed	None		X
	IT	HabBenBHT	None	X	
		PresPhyLoss	None	X	
		PresPhyDisturbSeabed	None		X
	MT	PresPhyLoss	None	X	
		PresPhyDisturbSeabed	None		X
	SI	PresPhyLoss	None	X	
		PresPhyDisturbSeabed	None		X

Region	**MS**	**Feature**	**Element**	**D6C3**	**D6C4**	**D6C5**
BAL	DE	HabBenBHT	Infralittoral rock and biogenic reef	X	X	X
			Infralittoral mixed sediment	X	X	X
			Infralittoral sand	X	X	X

Region	MS	Feature	Element	D6C3	D6C4	D6C5
			Infralittoral mud	X	X	X
			Circalittoral rock and biogenic reef	X	X	X
			Circalittoral mixed sediment	X	X	X
			Circalittoral sand	X	X	X
			Circalittoral mud	X	X	X
	DK	HabBenBHT	Infralittoral rock and biogenic reef	X	X	
			Infralittoral coarse sediment	X	X	
			Infralittoral mixed sediment	X	X	
			Infralittoral sand	X	X	
			Infralittoral mud	X	X	
			Circalittoral rock and biogenic reef	X	X	
			Circalittoral coarse sediment	X	X	
			Circalittoral mixed sediment	X	X	
			Circalittoral sand	X	X	
			Circalittoral mud	X	X	
			Offshore circalittoral coarse sediment	X	X	
			Offshore circalittoral mixed sediment	X	X	
			Offshore circalittoral sand	X	X	
			Offshore circalittoral mud	X	X	
	FI	HabBenBHT	Infralittoral rock and biogenic reef			X
			Infralittoral coarse sediment			X
			Infralittoral mixed sediment			X
			Infralittoral sand			X
			Infralittoral mud			X
			Circalittoral rock and biogenic reef			x
			Circalittoral coarse sediment			X
			Circalittoral mixed sediment			X
			Circalittoral sand			X
			Circalittoral mud			X
			Offshore circalittoral rock and biogenic			X
			Offshore circalittoral coarse sediment			X
			Offshore circalittoral mixed sediment			X
			Offshore circalittoral sand			X
			Offshore circalittoral mud			X
	LT	HabBenBHT	Infralittoral rock and biogenic reef	X		
			Infralittoral coarse sediment	X		
			Infralittoral mixed sediment	X		
			Infralittoral sand	X		
			Infralittoral mud	X		
			Circalittoral rock and biogenic reef	X		
			Circalittoral coarse sediment	X		
			Circalittoral mixed sediment	X		
			Circalittoral sand	X		
			Circalittoral mud	X		
			Offshore circalittoral mixed sediment	X		

Region	MS	Feature	Element	D6C3	D6C4	D6C5
			Offshore circalittoral sand	X		
			Offshore circalittoral mud	X		
	LV	HabBenBHT	Infralittoral coarse sediment			X
			Infralittoral mixed sediment			X
			Infralittoral sand	X	X	X
			Circalittoral mixed sediment			X
			Circalittoral sand			X
			Circalittoral mud			X
	PL	HabBenBHT	Littoral rock and biogenic reef			X
			Littoral sediment	X		X
			Infralittoral rock and biogenic reef			X
			Infralittoral mixed sediment	X		X
			Infralittoral sand	X		X
			Infralittoral mud	X		X
			Circalittoral sand	X		X
	SE	HabBenBHT	Infralittoral rock and biogenic reef	X	X	X
			Infralittoral sand	X	X	X
			Infralittoral mud	X	X	X
			Circalittoral rock and biogenic reef	X	X	X
			Circalittoral sand	X	X	X
			Circalittoral mud	X	X	X
			Offshore circalittoral rock and biogenic	X	X	X
			Offshore circalittoral sand	X	X	X
			Offshore circalittoral mud	X	X	X
NEA	BE	HabBenBHT	Infralittoral rock and biogenic reef	X		X
			Infralittoral coarse sediment	X		X
			Infralittoral sand	X		X
			Infralittoral mud	X		X
	DE	HabBenBHT	Littoral rock and biogenic reef	X		
			Littoral sediment	X	X	X
			Infralittoral sand	X	X	X
			Circalittoral coarse sediment	X	X	X
			Circalittoral sand	X	X	X
			Circalittoral mud	X	X	X
			Offshore circalittoral coarse sediment	X	X	X
			Offshore circalittoral sand	X	X	X
			Offshore circalittoral mud	X		X
	DK	HabBenBHT	Infralittoral rock and biogenic reef	X	X	
			Infralittoral coarse sediment	X	X	
			Infralittoral mixed sediment	X	X	
			Infralittoral mud	X	X	
			Infralittoral sand	X	X	
			Circalittoral rock and biogenic reef	X	X	
			Circalittoral coarse sediment	X	X	
			Circalittoral mixed sediment	X	X	

Region	MS	Feature	Element	D6C3	D6C4	D6C5
			Circalittoral sand	X	X	
			Circalittoral mud	X	X	
			Offshore circalittoral coarse sediment	X	X	
			Offshore circalittoral mixed sediment	X	X	
			Offshore circalittoral sand	X	X	
			Offshore circalittoral mud	X	X	X
			Upper bathyal rock and biogenic reef	X	X	
	ES	HabBenBHT	None	X	X	X
	FR	HabBenBHT	Infralittoral rock and biogenic reef	X		
			Infralittoral coarse sediment	X		
			Infralittoral mixed sediment	X		
			Infralittoral sand	X		
			Infralittoral mud	X		
			Circalittoral rock and biogenic reef	X		
			Circalittoral coarse sediment	X		
			Circalittoral mixed sediment	X		
			Circalittoral sand	X		
			Circalittoral mud	X		
			Offshore circalittoral rock and biogenic reef	X		
			Offshore circalittoral coarse sediment	X		
			Offshore circalittoral mixed sediment	X		
			Offshore circalittoral sand	X		
			Offshore circalittoral mud	X		
			Upper bathyal rock and biogenic reef	X		
			Upper bathyal sediment	X		
			Lower bathyal rock and biogenic reef	X		
			Lower bathyal sediment	X		
			Abyssal	X		
	IE	HabBenBHT	Benthic Broad Habitats*		X	X
	NL	HabBenBHT	Circalittoral coarse sediment	X		
			Circalittoral sand	X	X	
			Circalittoral mud	X		
			Offshore circalittoral coarse sediment	X		
			Offshore circalittoral sand	X		
			Offshore circalittoral mud	X		
	PT	HabBenBHT	Infralittoral rock and biogenic reef	X	X	X
			Infralittoral mixed sediment	X	X	X
			Infralittoral sand	X	X	X
			Infralittoral mud	X	X	X
			Circalittoral rock and biogenic reef	X	X	X
			Circalittoral coarse sediment	X	X	X
			Circalittoral mixed sediment	X	X	X
			Circalittoral sand	X	X	X
			Circalittoral mud	X	X	X
			Offshore circalittoral rock and biogenic	X	X	X

Region	MS	Feature	Element	D6C3	D6C4	D6C5
			Offshore circalittoral coarse sediment	X	X	X
			Offshore circalittoral mixed sediment	X	X	X
			Offshore circalittoral sand	X	X	X
			Offshore circalittoral mud	X	X	X
			Upper bathyal rock and biogenic reef	X	X	X
			Upper bathyal sediment	X	X	X
			Lower bathyal rock and biogenic reef	X	X	X
			Lower bathyal sediment	X	X	X
			Abyssal	X	X	X
	SE	HabBenBHT	Infralittoral rock and biogenic reef	X	X	X
			Infralittoral coarse sediment	X		
			Infralittoral mixed sediment	X		
			Infralittoral sand			X
			Infralittoral mud			X
			Circalittoral rock and biogenic reef			X
			Circalittoral coarse sediment	X		
			Circalittoral mixed sediment	X		
			Circalittoral sand			X
			Circalittoral mud			X
			Offshore circalittoral rock and biogenic			X
			Offshore circalittoral coarse sediment	X		
			Offshore circalittoral mixed sediment	X		
			Offshore circalittoral sand	X		X
			Offshore circalittoral mud			X
			Upper bathyal sediment	X		
MED	CY	HabBenBHT	Offshore circalittoral rock and biogenic	X		
	ES	HabBenBHT	None	X	X	X
	FR	HabBenBHT	Infralittoral rock and biogenic reef	X		
			Infralittoral coarse sediment	X		
			Infralittoral mixed sediment	X		
			Infralittoral sand	X		
			Infralittoral mud	X		
			Circalittoral rock and biogenic reef	X		
			Circalittoral coarse sediment	X		
			Circalittoral sand	X		
			Circalittoral mud	X		
			Offshore circalittoral rock and biogenic reef	X		
			Offshore circalittoral coarse sediment	X		
			Offshore circalittoral sand	X		
			Offshore circalittoral mud	X		
			Upper bathyal rock and biogenic reef	X		
			Upper bathyal sediment	X		
			Lower bathyal rock and biogenic reef	X		
			Lower bathyal sediment	X		
			Abyssal	X		

Region	MS	Feature	Element	D6C3	D6C4	D6C5
	HR	HabBenBHT	Infralittoral rock and biogenic reef	X	X	X
			Infralittoral mixed sediment	X	X	X
	MT	HabBenBHT	Littoral rock and biogenic reef	X	X	X
			Infralittoral rock and biogenic reef	X	X	X
			Infralittoral sand	X	X	X
			Circalittoral coarse sediment	X	X	X
			Upper bathyal rock and biogenic reef	X	X	X
			Upper bathyal sediment	X	X	X
			Posidonia oceanica meadows**	X	X	X
	SI	HabBenBHT	Littoral rock and biogenic reef	X	X	X
			Littoral sediment	X	X	X
			Infralittoral mixed sediment	X	X	X
			Infralittoral rock and biogenic reef	X	X	X
			Circalittoral rock and biogenic reef	X	X	X
			Circalittoral coarse sediment	X	X	X
			Circalittoral mud	X	X	X
BLK	RO	HabBenBHT	Littoral rock and biogenic reef	X	X	X
			Littoral sediment	X	X	X
			Infralittoral rock and biogenic reef	X		X
			Infralittoral sand	X	X	X
			Infralittoral mud	X		X
			Circalittoral rock and biogenic reef	X	X	X
			Circalittoral coarse sediment	X		
			Circalittoral mixed sediment	X		
			Circalittoral sand	X	X	X
			Circalittoral mud	X	X	X
			Offshore circalittoral sand	X		
			Offshore circalittoral mud	X	X	X

(*) IE grouped all the BHT in a single element included manually in the field.
(**) Reported by MT despite this element is not a BHT.

Findings: 16 MS of the overall 20 reported elements for D6C1 (extent of physical loss of seabed) and D6C2 (extent of physical disturbance of seabed). Of these, only five MS provided a complete assessment for at least one of the D6 criteria (LV, PL, IE, NL, MT).

Some MS did not follow the specification in the GES Decision (EU, 2017) to develop D6C3 assessment based on D6C2 (HR, LT, RO, SE) or the assessment of D6C4 based on D6C1 (HR, RO, SE), or did not use D6C1 and D6C2 (FI, LT, RO, SE). LT assessed only criterion D6C3 and considered the outcome as the final GES status for D6 habitats, while FI assessed only D6C5 by using the data from its national assessment report. Similarly, some MS assessed the extent of the loss of seabed (D6C1), without assessing the loss per habitat type under D6C4 (IE).

DE-BAL did not assess D6C1 and D6C2 because of lack of regional agreement: “The HELCOM-developed indicator (Cumulative Impact on benthic biotopes) for the criteria D6C1, D6C2, D6C3 could not be used due to a lack of regionally agreed thresholds. Therefore, only a descriptive description based on the Baltic Sea Pressure and Impact Index has been included in the state analysis.”. Similarly, SE-BAL reported that: “Sweden do not have indicators under this criterion yet. The reason is lack of enough and adequate information and knowledge on the benthic habitats. We are active in groups discussing this in CIS-MSFD, as well as in

HELCOM. A justification for not using criterion D6C3 is included in ART9GES-Schema for the Baltic.".

Gaps: Some MS did not assess any element (e.g., HR, PT). Lack of coordination in the way the MS assessed and reported the five criteria and in the way they combined them.

Recommendation for reporting: All D6 criteria are primary and all of them were expected to be equally reported by MS. For criteria D6C3, D6C4 and D6C5 there is no consistency in the way they were reported, since some MS reported all, while others less and the selection was not always made based on the available assessments. TG Seabed can develop a common understanding and clarifications in the reporting guidance on how the D6 criteria are linked together and how to report the available assessments across regions in a consistent way.

Recommendation for implementation: D6C1 and D6C2 were reported by 16 out of the 20 MS, and assessed only by five MS. Although there is some consistency in the way these two criteria were reported, lack of data seems to prevent a regional overview of these physical pressures acting on the seabed. Lack of data and thresholds led to a reduced number of assessments for criteria D6C3, D6C4 and D6C5 compared with the number of habitats reported as being present in the MS's waters. Moreover, for these criteria there is no consistency or pattern reported or assessed per MS and at regional level. Some MS reported criteria without providing assessments, while others reported only criteria for which assessments were conducted. In the second case, it was observed that the assessments are not always comparable or based on the same logic. Since available data exist, TG Seabed can support regional coordination, considering the work of the RSC, and develop further threshold values to harmonise the assessment of criteria across MS.

2.2.4 Legislation related to sea-floor integrity at EU and regional levels: reuse of other Directives or RSC assessments

Some MS specified in the text fields of the assessment that they used other Directives/EU polices (e.g., WFD, HD) or assessments from other MSFD descriptors for the D6 assessment. The Article 8 guidance (Walmsley et al, 2017) also recommends assessing criterion D6C5 considering the outcome of the assessment of D6C3 and other descriptors (e.g., D2, D5, D7, D8).

Regarding the assessments, some MS built them on EU Directives or RCS assessments:

- WFD used by CY, DE, DK, FI, PL, HR, MT (the last two based on the CARLIT - Cartography of Littoral - method);
- HD used by DE, DK, FI, SE;
- HELCOM used by DE; and
- OSPAR used by DE, DK, NL, BE.

FI and DK used indicators from the HD to assess D6C5. DK was the only MS to use an indicator from the WFD, 'DK-WFDEELFAUNA'.

BE, DK, DE and NL used the OSPAR indicators 'OSPAR-PHYS_DAM' for D6C2 and D6C3, and 'OSPAR-COND_BENT_HAB', 'ANS-OSPAR-BUND', and 'ANSNL-OSPAR-D6C5' for D6C5'. In some cases, as of DE, the indicator was associated with a national threshold value. Despite reporting the use of other assessments for the evaluation of the status of BHT or other habitat types for

D6, in many cases the parameters were reported as not assessed, without specifying any additional information on the outcome of the original assessment.

2.3 Completeness of reported MSFD methodology level

The presence of all components of the BHT assessment methodology was checked against the progress of the assessment and the use of criteria.

Data: Member States 2018 reports (file in Annex: D6-Art.8, columns: parameter, threshold value, threshold value source, trend).

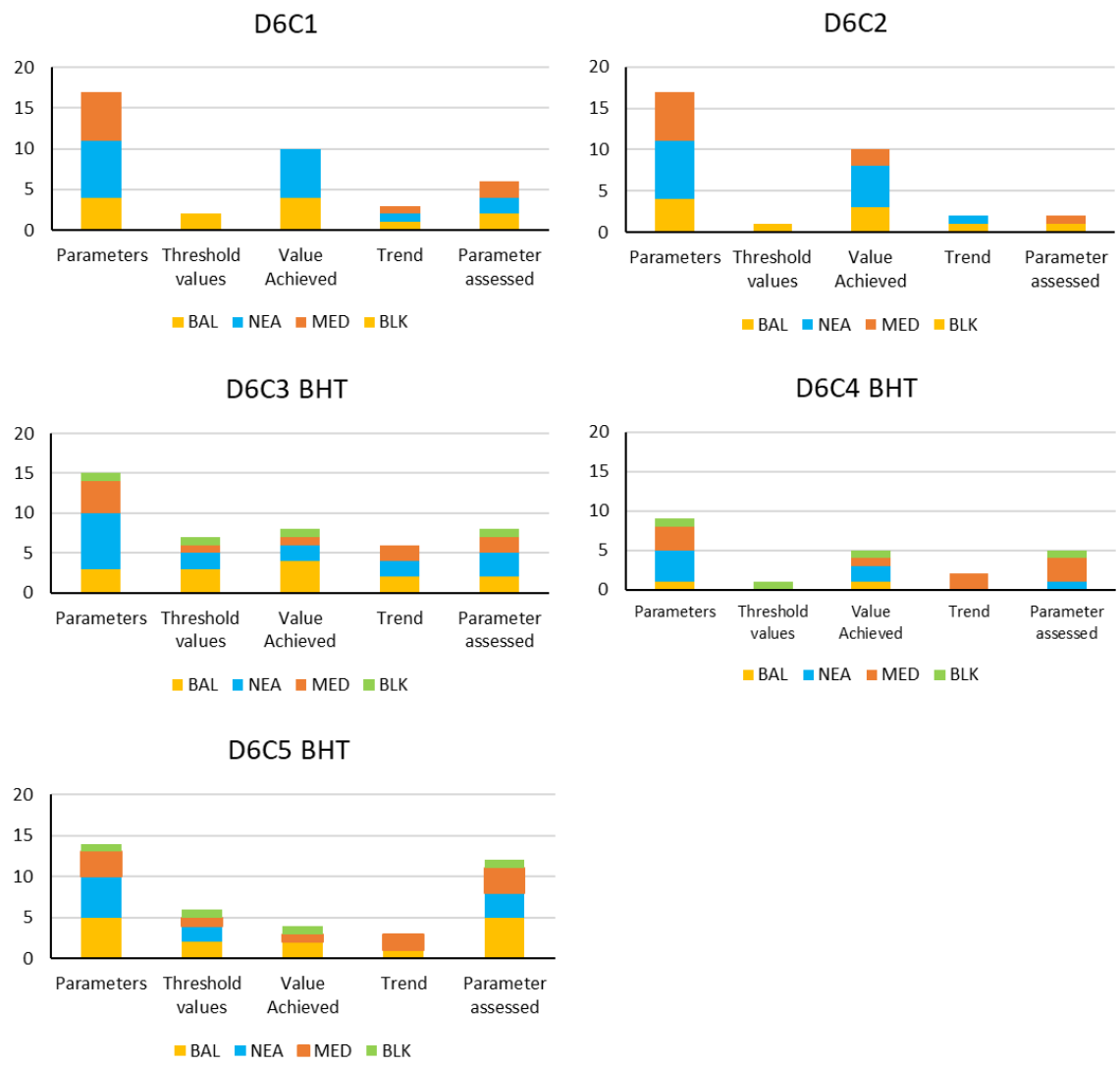

Figure 1. Number of MS by region who reported parameters, threshold values, values achieved, trends and parameter assessed (latter with a conclusive outcome: yes or no) for BHT in D6.

Findings: For D6C1 and D6C2, most MS reported the parameters used for the assessment, while the threshold values were not required for these criteria, although some MS (LV, PL) reported them.

For D6C5, few MS reported aspects of the assessment methodology, and many MS reported a conclusion for the parameter assessment based on other Directive's assessments (e.g., Art.17 of the HD), often without providing details of the assessment methodology (e.g., DE and CY).

DK provides all aspects of the assessment methodology but without a conclusion in the 'Parameter achieved', which they justified because of the lack of agreed threshold values at EU or RSC level.

Parameters were reported for all criteria and more details are given in the section 2.3.1. The number of MS reporting threshold values varies across the criteria; seven MS reported thresholds for D6C3 and six for D6C5, the two criteria that had the most MS reporting threshold values. Achieved values were reported mostly for the pressure criteria D6C1 and D6C2 and the impact criterion D6C3 (ten, ten and eight MS respectively) across all regions. Very few MS reported trends, which were mostly for criterion D6C3 (six MS across all regions). The number of MS reporting a conclusive outcome (either yes or no) in the parameter achieved field also differs across the five criteria. Moreover, the MS used inconsistent ways to report the achievement of parameters, for instance, some MS considered a parameter assessed when an achieved value is reported, while others when only a threshold value is reported. Many MS who provided achieved values reported the status of the parameter as unknown.

Gaps: Trends misreported in 'Parameter achieved' column (DE), result reported in 'Parameter achieved' (FR); both with final conclusions because uncertainty of the data.

The main reasons reported by the MS for not providing any assessment were missing threshold values at EU level, missing data (e.g., on small-scale fisheries) do not allow evaluation of element/criteria, and not enough information to provide harmonised and coherent assessments. Figure 1 is indicative for the missing information required for a complete assessment for the five criteria of D6.

Recommendation for reporting: There are more cases of misreported information, e.g., trends were reported under other fields, extent of loss, disturbance or adverse effect are reported in the 'Achieved values' field or in the text fields instead of in the 'Proportion value' field. This leaded to more situations where the extent of loss/disturbance/impact could not be automatically extracted from the reports (e.g., to provide an aggregate view per region or subregion).

Recommendation for implementation: The numerous gaps in the required assessment information (parameter, value, threshold, status of the parameter) did not allow for an overview of the status of the benthic habitats. In addition, MS' assessed parameters were reported in an inconsistent manner leading to different results, even in the cases of similar data availability. The data and methodological gaps have led to very inconsistent reporting by MS. TG Seabed should develop a common understanding on how to assess and report the benthic habitats' assessments, through the Art. 8 guidance, and MS should streamline the data and methodological gaps through the updates of the monitoring programmes, pursuant to MSFD Art. 11.

2.3.1 Parameters

We analysed the parameters reported per region (Table 7 and Table 8), which largely come from a predefined list in the reporting guidance. The outcome provides a good indication of the variety of information reported, the consistency and the areas where refinement is required. Table 7 presents the use of the default parameter in the assessment of D6 criteria. The colours correspond to the criteria: yellow (D6C1), orange (D6C2), red (D6C3), green (C4), blue (D6C5). The number of different parameters reported as "other" (those that are not included in the drop-down menu of the template) is shown in the cells in the last row and is displayed by criterion and region.

Data: Member States 2018 reports (file in Annex: D6-Art.8; columns: parameter, other parameter, threshold value, trend).

Table 7. Use of the default parameter in the assessment of D6 criteria. Colours: yellow (D6C1), orange (D6C2), red (D6C3), green (C4), blue (D6C5).

Parameters	BAL					NEA					MED					BLK				
EXT																				
HAB-CON																				
Other *	1	1	3		2			4		29			1		4			2	1	2

*List of the other (OTH) parameters in Table 8

Table 8. Other parameters ('Other' in Table 7) used by the MS and reported in the 'Other parameter' field.

Criteria	Other parameters	Regions			
		BAL	NEA	MED	BLK
D6C3	Distribution EUNIS habitats level 3		BE		
	Extent		DE		
	Extent? not affected		DE		
	Extent achieving Threshold for Good/High Status			MT	
	Extent of bottom Trawling	LT			
	Extent of sand excavation/dumping	LT			
	Habitat present, not assessed				RO
	M-AMBI* (n)				RO
	PL_WskZm	PL			
	Surface hard substrate vs soft substrate		BE		
	Not specified	DE	DE		
D6C4	Habitat present, not assessed				RO
	Not specified	DE	DE		
D6C5	Abundance of *Branchiostoma lanceolatum* in Nephtys-Ophelia habitat		BE		
	Abundance of *Echinocardium cordatum* in Abra habitat		BE		
	Abundance of *Echinocardium cordatum* in Nephtys-Ophelia habitat		BE		

Criteria	Other parameters	Regions			
		BAL	NEA	MED	BLK
	Abundance of *Lagis koreni* in Abra habitat		BE		
	Abundance of *Lanice conchilega* in Abra habitat		BE		
	Abundance of Mya spp. In Abra habitat		BE		
	Abundance of *Owenia fusiformis* in Abra habitat		BE		
	Abundance of Pestarella in Nephtys-Ophelia habitat		BE		
	Abundance of *Venerupis corrugata* in Abra habitat		BE		
	B-index	PL			
	Disturbed clusters of *Pomatoceros triqueter*		BE		
	Disturbed specimens of *Asteria rubens*		BE		
	EEI-c indeks			SI	
	Extent achieving Threshold for Good/High Status			MT	
	Habitat present, not assessed				RO
	Index	PL			
	M-AMBI indeks			SI	
	M-AMBI* (n)				RO
	Number of poseidon bunches (*Posidonia oceanica*)			SI	
	Relative occurrence of adult specimens of Alcyonidium spp.		BE		
	Relative occurrence of adult specimens of *Alcyonium digitatum*		BE		
	Relative occurrence of adult specimens of *Flustra foliacea*		BE		
	Relative occurrence of adult specimens of *Haliclona oculata*		BE		
	Relative occurrence of adult specimens of *Mytilus edulis*		BE		
	Relative occurrence of adult specimens of *Ostrea edulis*		BE		
	Relative occurrence of full grown reefs of *Buccinum undatum*		BE		
	Relative occurrence of full grown reefs of *Sabellaria spinulosa*		BE		
	Species Richness among Ascidiacea		BE		
	Species Richness among Bivalvia		BE		
	Species Richness among Bryozoa		BE		
	Species Richness among Cnidaria		BE		
	Species Richness among Echinodermata		BE		
	Species Richness among Gastropoda		BE		
	Species Richness among Malacostraca		BE		
	Species Richness among Maxyllopoda		BE		
	Species Richness among Polychaeta		BE		
	Species Richness among Porifera		BE		

Criteria	Other parameters	Regions			
		BAL	NEA	MED	BLK
	Not specified	DE	DE		

Findings: The parameter EXT (benthic habitat extent) was used to assess all D6 criteria in BAL and NEA, to assess D6C1, D6C2, D6C3 and D6C4 in the MED and to assess D6C3 and D6C4 in BLK, while HAB-CON (benthic habitat condition) was the default parameter for all region to assess D6C5. PL used an additional national parameter in the assessment of D6C1 and D6C2.

Another 44 parameters (OTH) were used to assess D6C3 (9 parameters), D6C4 (1 parameter) and D6C5 (37 parameters), some shared between more criteria. Some of these are repeats or refinements of the two main parameters (extent and condition). DE did not report the parameter used for the assessment. RO reported to have used another parameter to assess D6C1, D6C2 and D6C3, while for the assessment of D6C3, D6C4 and D6C5 reported "Habitat present, not assessed".

ES, LV and PT did not report any parameter for D6C2, D6C3 and D6C4. The main reasons for not reporting the parameter were lack of regional agreements (DE-HELCOM), lack of data (PT), results from other projects not yet available (PT-AMA) and no indicators for the criterion (SE-ANS).

Gaps: Lack of common assessment methods to harmonise the parameters at regional level, and at the habitat level mostly for D6C5.

Recommendation for reporting: There are several cases where the MS reported as "other" parameters that could fit to the default categories, e.g., "extent" was reported as "other" instead of "EXT". The availability in 2018 reporting of drop-down menus with a predefined set of parameters had greatly improved the consistency in the reporting compared to the 2012 reporting, however the list should be further revised according to the MS reported information. The reporting guidance should be updated, accordingly.

Recommendation for implementation: The development of common regional methodologies and the developing work within TG Seabed will facilitate the harmonisation of the parameters reported at regional level and across the same criteria, especially for D6C5.

2.3.2 Threshold values for sea-floor integrity/benthic habitats and their sources

This analysis gives an overview of the reported threshold values and of their sources among national, EU and RSCs for the broad habitat types and other habitat types (Table 9). Threshold values are not yet established at EU and regional level for D6, although TG Seabed is tasked with proposing EU values. TG Seabed encouraged the MS to use assessment for other Directives, i.e., WFD and HD, for the assessment of D6C5.

Data: Member States 2018 reports (file in Annex: D6-Art.8; columns: threshold value upper, threshold value source, feature, element).

Table 9. Overview of the threshold values reported and their sources per criterion and habitat type (elements). Elements were specified only for BHT.

Criteria	Threshold value	Units	Source	Parameter	Feature	Element	MS
D6C1	97.0	%	National	EXT	HabBenBHT		LV
D6C1 D6C2 D6C3	10.0	%	National	PL_WskZm	HabBenBHT	Infralittoral mud Infralittoral sand Infralittoral mixed sediment Littoral sediment Circalittoral sand	PL
D6C3	1.0		WDF	Extent achieving Threshold for Good/High Status	HabBenBHT	Infralittoral rock and biogenic reef Infralittoral sand Littoral rock and biogenic reef	MT
D6C3	0.68	ratio	(Sub) regional	M-AMBI* (n)	HabBenBHT	Circalittoral mud Circalittoral sand Infralittoral sand	RO
D6C3	10.0	%	National	EXT	HabBenOther		EE
D6C3	1.0 (Trend as GES definition)	Trend indicator	National	HAB-CON	HabBenBHT HabBenOther	Circalittoral coarse sediment Circalittoral mud Circalittoral sand Offshore circalittoral coarse sediment Offshore circalittoral mud Offshore circalittoral sand	NL
D6C3	25.0	-	National	Extent	HabBenBHT	Littoral sediment Circalittoral mud Circalittoral sand Infralittoral mud Infralittoral sand	DE
D6C3	10.0	-	National	Extent? not affected	HabBenBHT	Circalittoral coarse sediment Circalittoral mud Circalittoral sand Infralittoral sand Offshore circalittoral coarse sediment Offshore circalittoral mud Offshore circalittoral sand	DE
D6C3	0.61	ratio	National	M-AMBI*	HabBenBHT	Offshore circalittoral mud Circalittoral mud	RO

Criteria	Threshold value	Units	Source	Parameter	Feature	Element	MS
						Infralittoral sand	
D6C3 D6C5	0.68	ratio	National	M-AMBI*	HabBenBHT	Offshore circalittoral mud Circalittoral mud Infralittoral sand	RO
D6C3	>1.0 – < 0.55	EQR		EXT	HabBenBHT	Posidonia oceanica meadows	MT
D6C3	10.0	%		Extent of bottom Trawling	HabBenBHT	Circalittoral habitats Offshore habitats	LT
D6C3	1.0	%		Extent of sand excavation/dumping	HabBenBHT	Circalittoral habitats Infralittoral habitats Offshore habitats	LT
D6C4	5.0	Km2 ratio	(Sub) regional National	EXT	HabBenBHT	Circalittoral sand Infralittoral sand Circalittoral mud Offshore circalittoral mud	RO
D6C4	5.0	%	National	EXT	HabBenOther		EE
D6C4	-1.0	%		Not specified	HabBenBHT	Infralittoral rock and biogenic reef	DE
D6C5	0.6	index	HELCOM BEAT based national application	HAB-CON	HabBenOther		EE
D6C5	1.0	index	WDF	HAB-CON	HabBenBHT	Infralittoral rock and biogenic reef Infralittoral sand Littoral rock and biogenic reef Posidonia oceanica meadows	MT
D6C5	0.8	EQR	WDF	HAB-CON	HabBenBHT	Littoral rock and biogenic reef Infralittoral sand Infralittoral mixed sediment Infralittoral rock and biogenic reef	PL
D6C5	3.18	index	WDF	HAB-CON	HabBenBHT	Infralittoral sand Infralittoral mud Infralittoral mixed sediment Circalittoral sand Littoral sediment	PL

Criteria	Threshold value	Units	Source	Parameter	Feature	Element	MS
D6C5	0.123	index	WDF	HAB-CON	HabBenBHT	Infralittoral sand	PL
D6C5	0.6	-	National	HAB-CON	HabBenOther		EE
D6C5	6.0	index	National	HAB-CON	HabBenBHT HabBenOther	Infralittoral mud Infralittoral rock and biogenic reef	RO
D6C5	25.0	dmnl	National	HAB-CON	HabBenBHT	Littoral sediment Circalittoral mud Circalittoral sand Infralittoral mud Infralittoral sand	DE
D6C5	1.0	%	National	HAB-CON			NL
D6C5	0.6	Trend	EU*	HAB-CON	HabBenBHT	Infralittoral coarse sediment Infralittoral mud Infralittoral sand	BE

(*) Commission Decision 2008/915 / EC for EQR and qualitative (no threshold) for Margalef Diversity Index for Fisheries Assessment

Table 10. Quantitative threshold values used to assess D6 and their sources by MS and region.

Criteria	Qualitative threshold	Source	Parameter	MS
D6C1 D6C2	Criterion value 0-1		EXT	CY
D6C3	Disturbance category lower than 4	National	HAB-CON	SE
D6C3	Stable within accuracy margin	National	Distribution EUNIS habitats level 3	BE
D6C3	No negative trend	National	Surface hard substrate vs soft substrate	BE
D6C4	The habitat must be maintained at the current extent	National	EXT	IT
D6C4	At least 10% of the areas currently exploitable by benthic impacting fishing gears must not suffer of further adverse impacts	National	EXT	IT
D6C4	Trend in habitat extent is stable		EXT	MT
D6C4	Based on expert Judgement		EXT	IT

Findings: Although no threshold value was required for D6C1 and D6C2, LV and PL specified a value.

Four MS reported qualitative threshold values (Table 10). For D6C3, D6C4 and D6C5, five MS reported to use trends for the assessment, while the other reported thresholds are a mixture of 'extent' thresholds (%) and 'quality' thresholds (usually an EQR value). About the 19% of the threshold values in the assessments do not have a reported source. In some cases, the source can be extracted from the text in the field 'Description parameter', where it was also specified if the elements were assessed by another Directive (WDF, HD) or by the RSCs.

Gaps: For several BHTs threshold values do not exist, and the lack of data prevents any attempt to establish such values. The use of threshold values from the HD or WFD cannot be fully evaluated as many thresholds are reported as national, even if they were developed for one of these policy requirements. No regional harmonization is evident in the regional sets of the reported thresholds.

Recommendation for reporting: Despite that no threshold value is required for D6C1 and D6C2, some MS reported thresholds. The thresholds extent should have been reported in the 'Proportion Threshold Value' field, however many MS reported it in the 'Threshold' field. The reporting tool can be stricter with the accepted values, or the reporting guidance can provide indicative values.

Recommendation for implementation: Generally, the MS reported available threshold values that were already developed by RSCs or for other EU policies. Given the lack of data and regional harmonisation, the reported threshold values can be a starting point for developing regional and EU thresholds. The MS should support their experts working regionally (for D6C3 thresholds) and contributing to TG Seabed (for D6C5 thresholds).

2.3.3 Trend analysis

TG Seabed states that for D6C3, D6C4 and D6C5, "in the absence of a known reference state and a quantitative threshold, an improving trend in quality can be used in the interim" (SEABED_3-2020-09 Adverse effects[3]). This information was not available when the MS reported under Art. 8 in 2018. Table 11 summarises the MS that reported trends per region, for each criterion and BHT.

Data: Member States 2018 reports (file in Annex: D6-Art.8; columns: trend).

Table 11. Trends reported by the MS and divided by criteria, BHT element and by region.

Criteria	Trend	BHT elements	BAL	NEA	MED
D6C1	Stable		PL	BE	MT
D6C2	Stable		PL	BE	
D6C3	Deteriorating	Circalittoral sand	LT		
D6C3	Stable	Abyssal		PT	
D6C3	Stable	Circalittoral coarse sediment	LT		
D6C3	Stable	Circalittoral mixed sediment	LT		
D6C3	Stable	Circalittoral mud	LT		
D6C3	Stable	Circalittoral rock and biogenic reef	LT		
D6C3	Deteriorating	Circalittoral sand	LT		
D6C3	Stable	Circalittoral sand	PL		

[3] https://circabc.europa.eu/w/browse/54beb651-0991-4fc2-8ceb-26fe3572f76f

Criteria	Trend	BHT elements	BAL	NEA	MED
D6C3	Stable	Infralittoral coarse sediment	LT		
D6C3	Stable	Infralittoral mixed sediment	LT PL		HR
D6C3	Stable	Infralittoral mud	LT PL		
D6C3	Improving	Infralittoral rock and biogenic reef			MT
D6C3	Stable	Infralittoral rock and biogenic reef	LT	BE	HR
D6C3	Stable	Infralittoral sand	LT PL		
D6C3	Improving	Littoral rock and biogenic reef			MT
D6C3	Stable	Littoral sediment	PL		
D6C3	Stable	Lower bathyal rock and biogenic reef		PT	
D6C3	Stable	Lower bathyal sediment		PT	
D6C3	Stable	Offshore circalittoral mixed sediment	LT		
D6C3	Stable	Offshore circalittoral mud	LT		
D6C3	Stable	Offshore circalittoral sand	LT		
D6C3	Stable	Posidonia oceanica meadows			MT
D6C3	Stable	Upper bathyal rock and biogenic reef		PT	
D6C3	Stable	Upper bathyal sediment		PT	
D6C4	Stable	Infralittoral mixed sediment			HR
D6C4	Stable	Infralittoral rock and biogenic reef			HR
D6C4	Stable	Littoral rock and biogenic reef			MT
D6C4	Stable	Posidonia oceanica meadows			MT
D6C5	Deteriorating	Infralittoral mud	PL		
D6C5	Improving	Infralittoral mud	PL		
D6C5	Improving	Infralittoral sand	PL		
D6C5	Improving	Littoral rock and biogenic reef	PL		MT
D6C5	Deteriorating	Littoral sediment	PL		
D6C5	Improving	Littoral sediment	PL		
D6C5	Improving	Infralittoral mixed sediment	PL		
D6C5	Stable	Infralittoral mixed sediment			HR
D6C5	Improving	Infralittoral rock and biogenic reef	PL		MT
D6C5	Stable	Infralittoral rock and biogenic reef			HR
D6C5	Stable	Posidonia oceanica meadows			MT

Findings: About 9% of all criteria and features (BHT + other benthic habitat types) reported were assessed using trends by five MS. The remaining 91% reported the status of trends as 'unknown' or 'not relevant' or did not report on trends at all. Regional consistency cannot be evaluated with such a small number of MS reporting on trends.

BE, IT, MT and NL reported in the field 'Threshold qualitative' to have used the trend for the assessment, although only BE and MT reported a trend status.

Gaps: The lack of time-series and data for the D6 criteria was evident in the very few MS that assessed criteria using trends. In some cases, this could be the consequence of the inability of MS to assess BHT (i.e., provide a status conclusion per criterion/parameter) in 2012 MSFD cycle, therefore they were not able to assess trends in status in 2018. However, in the HD the use of trend-based assessments is common, and we would assume that some time-series, or assessments could be directly picked-up from the HD.

Recommendation for reporting: No specific recommendations for the reporting, since the trend assessments were reported by all MS and this entry is very informative.

Recommendation for implementation: TG Seabed provided already some recommendation for the use of trends. It should be explored whether the assessments or the trends of the parameters from the HD could be better exploited for the MSFD.

2.3.4 Assessment of value unit used

For D6C1 and D6C2, the GES Decision recommends measuring the extent of the assessment area physically lost (D6C1) and physically disturbed (D6C2) in square kilometres (km^2), while for the extent of each habitat type adversely affected or lost (D6C3, D6C4 and D6C5) to use either in square kilometres (km^2) or as a proportion (percentage) of the total natural extent of the habitat in the assessment area.

This analysis summarises the reported value units (Table 12) for each criterion to identify the level of harmonisation across MS, especially at regional level, and to highlight potential discrepancies and their origin.

Data: Member States 2018 reports (file in Annex: D6-Art.8; columns: value unit, value unit other).

Table 12. Value units reported by the MS for the assessment of all criteria.

Criteria	Region	Value unit used		Other value unites used
		Km^2	%	
D6C1 D6C2	BAL	DK EE	PL LV	
	NEA	BE DK ES IE	NL	
	MED	ES SI		
	BLK			
D6C3	BAL	DK	PL	
	NEA	DK		
	MED	MT SI		
	BLK			RO - ratio
D6C4	BAL	DK	EE	
	NEA	DK IE	NL	
	MED	MT SI		
	BLK	RO		RO - ratio
D6C5	BAL			

NEA	IE	BE	BE - Species number
MED			SI - items/m2
BLK			RO - ratio RO - dmnl

Findings: To assess physical loss (D6C1) and disturbance (D6C2), six MS (30%) used square kilometres, while three MS (15%) used proportion of the total natural extent.

For the assessment of D6C3, four MS used km^2, and five MS used the percentage of the extent. RO used different value units for criterion D6C4.

Some MS (LV and NL) reported the value unit for D6C3 in the field 'Other value unit' although they used the default value unit, which is a reporting issue.

It should be noted that further clarification is needed in the monitoring guidance to harmonise the reporting of the MS. This derives from the fact that the units of Km^2 or % should be applied to the 'ProportionValueAchieved' field to express the extent of loss/disturbance/impacts and not to the 'ValueAchieved' field which is intended to express the quality of the habitat.

Because of the use of two different units for D6C1 and D6C2 in BAL and NEA, the data provided for the same BHT are not directly comparable. Apart from the need to report these extent values in the correct field, it is also apparent that both values should be reported for all criteria to get compatible data per criteria/BHT. The analysis of the reported units indicated lack of regional pattern or harmonisation, inconsistency in the reporting and the fact the MS reported what was available in terms of data and assessments. TG Seabed's work will help to harmonise assessments and reporting through a common understanding and an assessment guidance for Art. 8.

2.4 Consistency of assessment period

The MS reported the assessment period for each parameter per habitat and criterion. We analysed this information to explore potential discrepancies in the assessment at regional and criterion level that could potentially affect the comparability and the consistency in the final assessment of each parameter (Table 13).

Data: Member States 2018 reports (file in Annex: D6-Art.8; columns: assessment period, criteria).

Table 13. Assessment period for each criterion by region based on the number of parameters assessed.

	1980-2016	1983-2016	2007-2012	2007-2015	2008-2017	2010-2015	2010-2016	2011-2015	2011-2016	2012-2016	2012-2017	2012-2018	2013-2018	2014-2018	2015-2015	2016
Number of years	17	34	6	9	10	6	7	6	7	6	7	8	6	5	1	1
BAL																
D6C1		1						1	5						19*	
D6C2								1	5	1					19*	
D6C3	14			13					43		26					
D6C4	14			11					24							
D6C5				11	7				14 2							
NEA																
D6C1		1				1			4		5			1	1	2
D6C2		1				3			4		4			1	1	
D6C3	15		4	16			4		9		26 5	3			8	
D6C4	15		2	7		1			1		19 3	3			1	
D6C5	1		2	8	7	1	32		9		19 3	3			5	
None											1			22		
MED																
D6C1									2		8	2				
D6C2									2		5	2				
D6C3											40	10	2			
D6C4											22	9	2			
D6C5											23	9	2			
None												15	16			
BLK																
D6C3											21					
D6C4											7					
D6C5											15					

(*)MS reported as 2015-2015, here placed in the 2015 field to ease comparison with other MS.

Findings: The GES Decision foresees 6-year cycles of assessments, however it becomes clear from Table 13 that the reported period varies from the 1 to 34 years. As such, the starting period of the assessment varies accordingly, not generating comparable assessments. Even the end-dates of the assessments are not harmonised varying from 2012 to 2018 and deviating

from the MSFD cycle. One MS reported only the final year of the assessment as unique year of assessment, without specifying the duration of the period.

The 6-year cycle was not reflected by many MS' assessments for all criteria, while the assessment period ranged between 4 and 7 years.

Gaps: Obviously, the MS used what was available in terms of data, indicators, and assessments instead of adjusting them to the MSFD requirements – i.e., a 6-year cycle in this case. Of course, longer data sets can provide more accurate estimates, however, in most of the periods the duration is shorter than six years. The different assessment periods might not provide comparable assessments and might not reflect the same volume of pressures to the habitats.

In principle, the Art. 8 assessment is providing an assessment of status for the 6-year period and the trend in status compared to the previous 6-year period, but there is need to use data over a longer period when it is available, such as:

- Assessing total loss, essentially since anthropogenic pressures (infrastructures) began.
- Setting reference values based on a date when habitat quality was considered unimpacted.
- Assessing extent of adverse effect, when some of this impact could have occurred prior to the 6-year period but needs to be included in the assessment because the habitat is still impacted.

While the reporting structure is built to include all the above information, their reporting (when available) was not consistent across the MS.

Recommendation for reporting: The assessment period is crucial and should remain in the report. The reporting guidance and the Art. 8 guidance need improvements to reflect the issues above.

Recommendation for implementation: The different assessment period should be harmonised (as well as the understanding of what should be assessed and reported) to avoid producing incomparable and inconsistent environmental assessments. The alignment of the HD assessments with the MSFD could harmonize the assessment periods and allow the reuse of the HD assessments directly for the MSFD.

2.5 Targeted pressures

Figure 2 summarises the pressures assigned by the MS to each reported entry for the BHT for the three benthic habitat state criteria. A more detailed analysis of the pressures reported for each feature per MSFD region is given in Table 14.

Data: Member States 2018 reports (file in Annex: D6-Art.8; columns: GES achieved, element, feature).

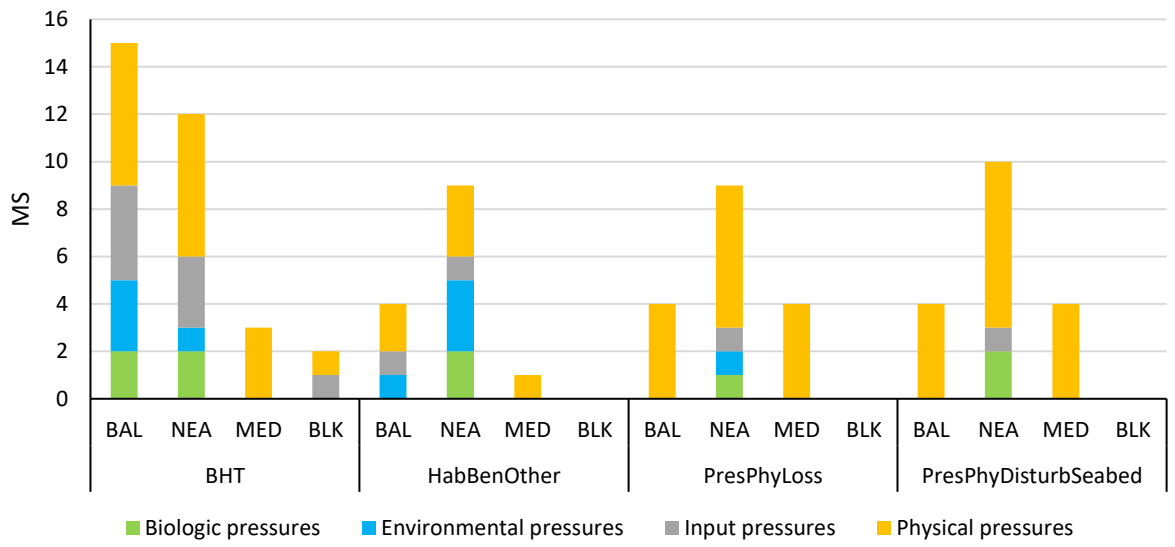

Figure 2. Number of MS per region reporting pressure groups by feature (BHT: broad habitat type, HabBenOther: other habitat types, PresPhyLoss: physical pressure, PresPhyDisturbSeabed: physical disturbance to seabed).

Table 14. Pressure groups and pressures by feature and region for D6 (BHT: broad habitat type, HabBenOther: other habitat types, PresPhyLoss: physical pressure, PresPhyDisturbSeabed: physical disturbance to seabed).

Pressure group	FEATURE			
Pressures	**HabBenBHT**	**HabBenOther**	**PresPhyLoss**	**PresPhyDisturbSeabed**
PresAll		BAL NEA BLK		
Biological pressures				
Disturbance of species	NEA			
Extraction of, or mortality/injury to, wild species	BAL NEA	NEA	NEA	NEA
Input of microbial pathogens	BAL			
Input or spread of non-indigenous species	BAL NEA			
Environmental pressures				
Eutrophication	BAL NEA	BAL NEA		
Hydrographical changes	BAL NEA	BAL NEA	NEA	
Litter in the environment		NEA		
Newly introduced non-indigenous species		NEA	NEA	
Input pressures				
PresInputAll	BLK			
Input of other substances	BAL NEA	BAL NEA	NEA	
Input of litter	BAL	NEA		NEA
Input of nutrients	BAL NEA	NEA	NEA	
Input of organic matter	BAL NEA	NEA	NEA	NEA
Physical pressures				
PresPhyAll	BAL BLK		BAL	
Physical disturbance to seabed	BAL NEA MED BLK	BAL NEA MED	BAL NEA MED	BAL NEA MED
Changes to hydrological conditions	BAL NEA	BAL NEA	BAL	BAL
Physical loss of the seabed	BAL NEA MED BLK	NEA MED	BAL NEA MED	BAL NEA MED

Recommendation for implementation: Very few MS reported pressures for the four features. Moreover, there is a mixture of different kind of pressures reported in each region, driven by the parameter, which is assessed, rather from the anthropogenic activity that need to be assessed and this can explain the different patterns across the regions. This could be caused by the structure of the reporting and could be clarified in the updated reporting guidance. The state-impact-pressure relationship should be better specified and strengthened, a recommendation which was also previously included in the in-depth assessment of the MS MSFD reports from 2012 (Palialexis et al. 2014). The habitats assessed by the state criteria of D6 could be monitored and analysed in line with the risk-based approach according to their exposure and sensitivity to pressures. However, the information extracted from the MS reports and presented in Figure 2 and Table 14 is very scarce and indicative, rather than comprehensive and informative for the key pressures in each MS and region. TG Seabed can support the development of a common understanding between state-impact-pressure for the benthic habitats and provide recommendations for harmonised assessments indicative to the prevailing regional pressures.

2.6 Date when GES is expected to be achieved

The MS reported for each feature when it is expected to reach GES. We analysed the reported information to get an overview of the date when the GES for each criterion is expected to be achieved at regional level (Table 15).

Data: Member States 2018 reports (file in Annex: D6-Art.8; columns: GES achieved, element, feature).

Table 15. Predicted date reported by MS by when GES is expected to be achieved for each D6 feature and associated criterion.

BAL		GES achieved	GES expected to be achieved			Not assessed	Not relevant	Unknown
			by 2020	later than 2020 (Art.14 exception)	later than 2020 (no Art.14)			
D6C1	HabBenBHT	LV						
	PresPhyDisturbSeabed						PL	
	PresPhyLoss	PL				DE PL	EE PL	DK
D6C2	HabBenBHT	LV						
	PresPhyDisturbSeabed	PL				DE	EE PL	DK
	PresPhyLoss						PL	
D6C3 D6C4 D6C5	HabBenBHT	LT LV PL	LV		DE LT PL	SE		DK FI

NEA		GES achieved	GES expected to be achieved			Not assessed	Not relevant	Unknown
			by 2020	later than 2020 (Art. 14)	later than 2020 (no Art.14)			
D6C1	PresPhyLoss	IE			NL	DE PT	BE FR	DK ES PT
D6C2	PresPhyDisturbSeabed				NL	DE PT	BE FR	DK ES IE
D6C3 D6C4 D6C5	HabBenBHT	IE			BE DE NL	ES PT SE		DK

MED		GES achieved	GES expected to be achieved			Not assessed	Not relevant	Unknown
			by 2020	later than 2020 (Art.14 exception)	later than 2020 (no Art.14)			
D6C1	HabBenBHT					IT		
	PresPhyLoss	CY			IT	SI	FR MT	ES
	PresPhyDisturbSeabed	CY			IT	SI	FR MT	ES
D6C3 D6C4	HabBenBHT					CY ES MT SI	FR	HR

BLK		GES achieved	GES expected to be achieved			Not assessed	Not relevant	Unknown
			by 2020	later than 2020 (Art.14 exception)	later than 2020 (no Art.14)			
D6C3 D6C4 D6C5	HabBenBHT				RO	RO		

Findings: Five MS reported to have achieved GES for some features, although no GES definition was provided in Art.9 and there is no regionally agreed GES determination. Some MS reported achieved GES based on national threshold values (PL, LT, LV, EE), expert judgement (IE) and estimation of low impact (CY, PT). In one case, a MS that applied the one-out all-out (OOAO) integration rule stated that the outcome for the criteria was turned from

positive to negative (not in GES) after the integration, as only one of its reporting areas (MRUs) achieved GES. 85% (17) of MS reported either 'not assessed/not relevant' or 'unknown'.

Gaps: Many gaps were noticed in data and methods, which led to inconsistent reporting for the date of GES achievement per feature. Since this entry requires consideration of several assessments and integrations, the outcome from Table 14 is not representative to the status of GES for BHT in the European waters. The MS had different understanding on how to assess or report GES achievement and followed different approaches. It is indicative that in each region each MS consistently reported one of the options for each criterion, however the same criterion and feature was differently reported across the regions. For instance, D6C1 in NEA was assessed by eight MS that provided five different outcomes: one MS achieved GES, one will achieve GES after 2020, two did not assess it, for two MS it was considered not relevant and for three the status is unknown. Considering that they refer to the same criterion for the same region, it seems contradictory to simultaneously report the above options. Thus, regional coordination is required for both assessment and reporting period, to tackle gaps and inconsistencies.

Recommendation for reporting: The MS need to combine assessments of other criteria and parameters to assess whether GES is achieved. This entry is reported at the BHT feature level and depends on integration of all parameters and criteria across all BHTs, however many MS conceived GES achievement differently and reported inconsistent information as described above in the gaps section, resulting in an inconsistent outcome by feature per region. This issue should be tackled in the reporting guidance, as well. MS seems to have used the categories 'Not assessed' and 'Unknown' in an interchangeable way to report not assessed elements; these categories could be merged in the reporting guidance. 'Not relevant' should be used only for D6C1 and D6C2 where no GES is expected to be achieved.

Recommendation for implementation: The reported information for this entry is the MS' estimation for when GES is expected to be achieved. The field 'GESextentAchieved 'should express the proportion of BHTs that are in GES (based on both quality and proportion thresholds). For D6 it is remarkable how few MS have attempted to report on this field or their ambition (GESextentThrehold). In addition, there are inconsistencies in how this field is reported, making it impossible to state how many BHTs are currently in GES. All the work that is related to the harmonisation of methodological standards (thresholds, integration rules) can ensure a consistent and comparable assessment of habitats in the coming cycles and achieve consistency in the MS GES assessment. The experts from the MS could develop this work through TG Seabed and this can inform the Art. 8 guidance accordingly. Contradicting assessments between MSFD, RSCs, and HD should be avoided.

2.7 Analysis of the levels of integration by parameter and criteria reported

The majority of the MS reported integration rules from different Directives or Descriptors, especially for D6C5. At the time of the reporting, in 2018, agreed integration rules were not available at regional scale. The OOAO rule for both parameter and criteria was reported by some MS (Table 16).

Data: Member States 2018 reports (file in Annex: D6-Art.8; columns: Integration rule type by parameter, Integration rule type by criteria, feature).

Table 16. Integration rule type by parameter (top) and by criteria (bottom).

Regions	Integration rule* type by parameter					Not relevant	Other
	NHIE_NWEI	NHIE_WEI	OOAO	SPATIAL	THRES		
BAL	LV	PL	LT	LV		DK EE PL SE	FI
NEA			DE			BE DK FR IE SE	BE NL
MED					SI	FR IT	
BLK			RO				

Regions	Integration rule* type by criteria			Not relevant	Other
	MULTIMETRIC	OOAO	SPATIAL		
BAL		EE LV	LV	DK EE PL SE	
NEA		DE	IE		NL
MED				FR	
BLK	RO	RO		RO	

(*) Integration rules reported: One-out-all-out (OOAO), Threshold methods (THRES), Non-hierarchical, non-weighted averaging (NHIE_NWEI), Non-hierarchical, weighted averaging (NHIE_WEI), Multi-metric indices (MULTIMETRIC), Spatial analysis (SPATIAL), and other integration methods (OTHER).

Findings: MS used integration rules in an inconsistent way, as shown in Table 16. For instance, the eight MS in the BAL reported six different options for the integration rules of the parameters. Considering the data gaps in the 2018 reporting, these inconsistencies could be tackled in the future through guidelines (e.g., Art. 8 guidance) for the appropriate D6 integration rule. Similarly, the integration of criteria should follow an agreed rule across all MS and in line with the overall GES assessment.

Gaps: Missing agreed integration rules. According to TG Seabed document (SEABED_3-2020-07_OtherPolicyMethods), these should be defined case by case.

Recommendation for reporting: An agreed integration method would facilitate the reporting for this entry and would provide an overview of comparable GES assessments across regions and habitats. When MS will reach an agreement of the integration rules for criteria and/or parameters these could be automatically derived in the reported tool.

Recommendation for implementation: The debate on which integration rules should be used is ongoing for several descriptors. For D6, TG Seabed is working to this direction and will continue testing the most promising methods.

3 Results of Art.9 analysis for D6 benthic habitats

3.1 Key results and recommendations from MSFD Article 9 reports

- All 20 MS included in the analysis reported for Art. 9 (eight MS in BAL, nine MS in NEA, seven MS in MED and one MS in BLK).
- 121 GES determinations were reported, at different levels (Descriptor, criterion) and specificity, with only a few detailed enough to support a quantitative GES determination.
- The many different GES determinations and the inconsistency in their reporting are caused by the lack of a common approach to determine GES and lack of data to support the GES determination. A common agreed GES determination, at least regionally, will harmonise or better lead to a single well understood quantitative determination for the same component and habitat type. RSCs can coordinate the work for a common and harmonised GES determination. It can also provide a clear case in the reporting guidance for the required information to be included in the GES determination field (e.g., elements to assess, parameter used and threshold value, integration type).
- Lack of data and agreed methods were the main reasons for not using a criterion or not setting threshold values for the assessment. Obviously, for the same reasons the MS could not provide quantitative determinations of GES and threshold values for the proportion of habitats impacted or quality threshold values to discriminate good from poor status.
- The GES determination should be harmonized at regional level in order to provide a comparable assessment for benthic habitats. The assessment flow and the Art. 8 guidance should be the basis for building a common GES determination in a consistent way and agreeing how the GES will be reflected at the level of broad habitat types, criteria and parameters/indicators. Since the MS have already invested considerable effort to harmonise methods for assessments, that has improved the common understanding of the MSFD requirements, the follow-up would be to use this common understanding for harmonising GES determination. Common integration rules and common agreed methods will bring consistency to all levels. The MSFD reporting tool (EEA's Reportnet 3.0) should be developed accordingly, linking GES determination to the proper level reducing and automatizing the reporting effort. Starting from the technical groups, the MS' appointed experts could propose how to formulate a common approach for a quantitative GES determination at regional or subregional level, which then could be endorsed by the MS through the MSFD CIS groups.

3.2 GES determination levels

The GES definitions reported by the MS were grouped according to the targeted feature (Figure 3) and classified according to the level where GES was determined (levels are in x-axis of Figure 4). In Figure 4, levels are determined as the most detailed information provided in the GES determination, as follows: a) for Descriptor level, when D6 definition is included; b) for criteria level, when GES Decision criteria definition is included; c) for Other Directive/RCS/national, when the determination includes reference to other legislation or agreements; d) for Indicator/threshold values (TV), when one or more elements used for the

GES assessment are included in the determination; e) not relevant level, when GES determination is not relevant for the reported feature.

Data: Member States 2018 reports (file in Annex: D6-Art.9; columns: GES definition, GES component, Determination date).

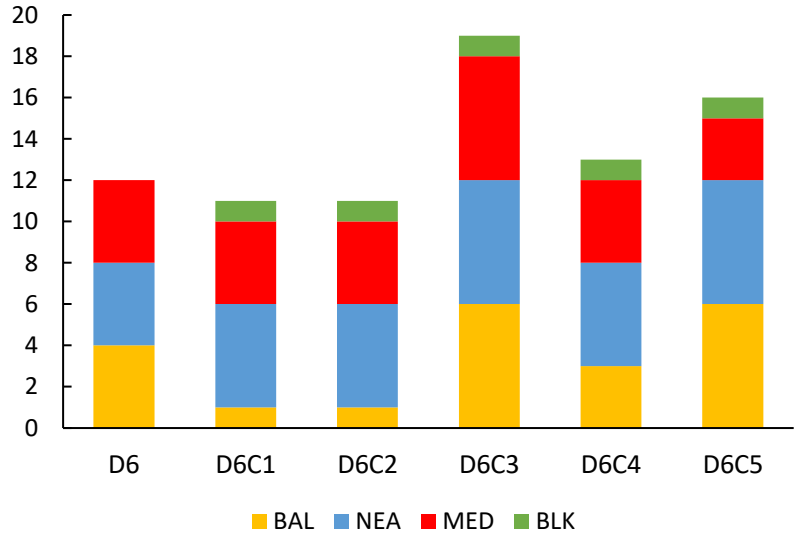

Figure 3. Number of GES determinations provided by MS by criterion.

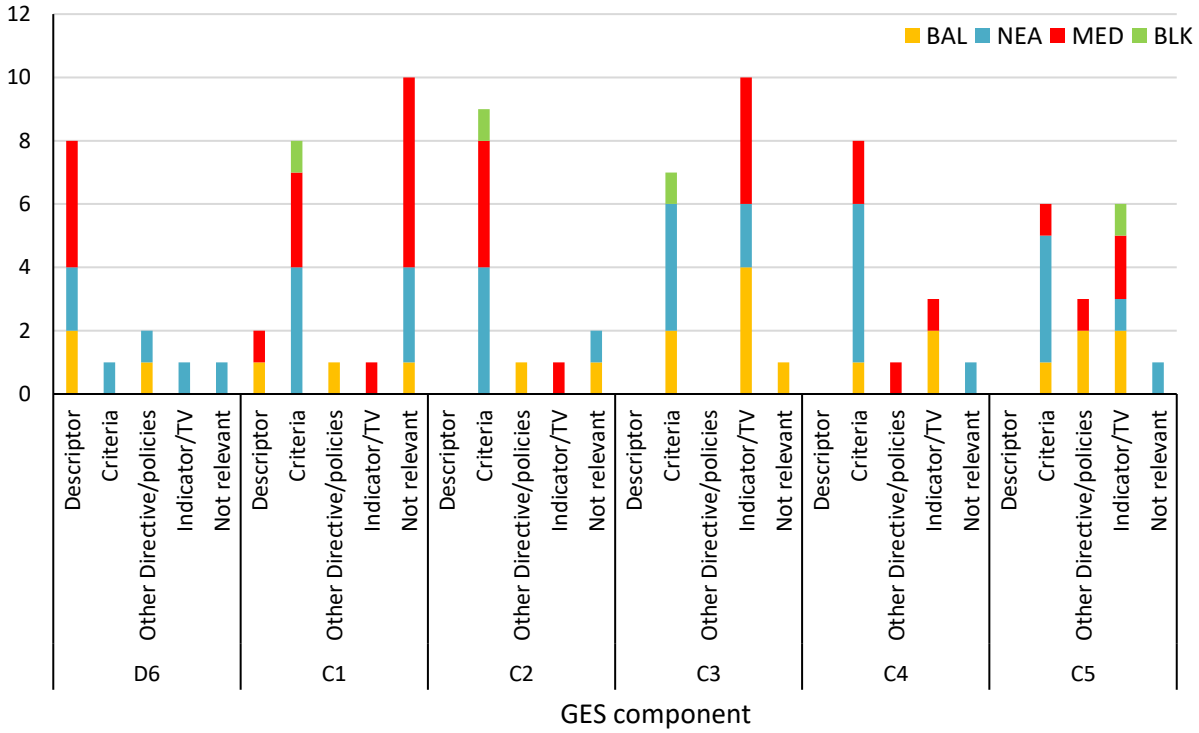

Figure 4. Analysis of the levels of GES determination reported by the MS, divided by criteria and region.

Findings: At the descriptor level for D6, four MS provided a single GES determination that was meant to cover all criteria. At both descriptor and criteria level, most of the MS provided generic GES determinations that repeated or rephrased the criteria definitions of the GES Decision.

Generally, the MS need to harmonise the level at which GES determination is reported. Many MS reported at the generic GES component (D6) and most of the definitions reiterate the GES Decision. For all criteria, very few GES determinations included more detailed description, like indicators, threshold values or other information. Many determinations are vague without providing a qualitative or even better a quantitative GES description. There is no harmonisation across the MS on how and at which level GES is determined and reported. Both issues are still under debate for most of the MSFD Descriptors.

Five MS in BAL and one in MED referred in their GES determination to other Directives or policies (WFD, HD, HELCOM, OSPAR). In the GES definition for D4C5, LT and SE reported to use indicators from other Descriptors (D5). SE used proxy for D6C5: "*proxies from other descriptors (D5) for the assessment, which includes assessments under Directive 2000/60/EC. In addition, we have used assessment from the Directive 92/43/EEC, but the assessment is not complete in relation to the broad habitat types*", while LT referred to another Descriptor's criterion: "*The species composition and relative abundance of macrofauna communities reach values indicating that the increase in nutrients and organic matter does not cause adverse effects. The indicator is assigned to criterion D5C8*."

11 MS provided GES determinations that included some aspect of the assessment methodology (either a threshold value, an indicator or an integration rule), while three MS (ES, NL, PT) reported to use trends for the criteria assessment.

Some MS reported in the GES determination field a justification for not providing a definition or the reason for the delay.

Gaps: Lack of harmonisation caused by lack of common understanding of the level at which GES should be determined and lack of a common approach to provide a qualitative GES determination at regional level.

Recommendation for reporting: This information is crucial and well-structured and explained in the reporting guidance. Given that most MS provided long descriptions for their GES determination (from a sentence to paragraph) it could be helpful if some of the reported GES characteristics are reported in separate categories. For instance, a category could be the qualitative or quantitative GES determination, and another one could be the units for the quantitative determination (e.g., percentage, ratio), always as a drop-down selection to facilitate reporting.

Recommendation for implementation: The assessment flow and the Art. 8 guidance should be the basis for building a common GES determination in a consistent way and agreeing how the GES will be reflected at the level of broad habitat types, criteria and parameters/indicators. Since the MS have already invested considerable effort to harmonise methods for assessments, that has improved the common understanding of the MSFD requirements, the follow-up would be to use this common understanding for harmonising GES determination. Common integration rules and common agreed methods will bring consistency in all levels. The MSFD reporting tool (EEA's Reportnet 3.0) should be developed accordingly, linking GES determination to the proper level reducing and automatizing the reporting effort. Starting from the technical groups, the MS' appointed experts could propose how to formulate

a common approach for a quantitative GES determination at regional or subregional level, which then could be endorsed by the MS through the MSFD CIS groups.

3.3 Analysis of adequacy of GES

Data: Member States 2018 reports (file in Annex: D6-Art.9; columns: Update type).

3.3.1 Level of ambition

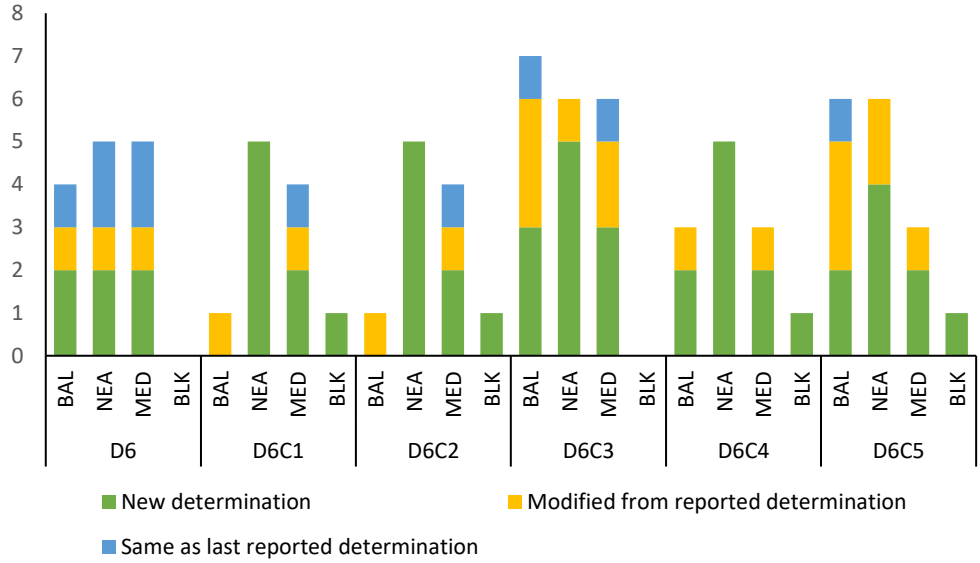

Figure 5. GES determinations update type in BAL, NEA, MED and BLK. Green: new determinations, Yellow: Modified from reported determination, Blue: Same as last reported determination.

Findings: Overall, 57.5% of the GES determinations are new (40% of those at Descriptor level, 68% for D6C1 and D6C2, 57% for D6C3, 69% for D6C4 and 46% for D6C5). 32.5% of GES determinations are modified from the last reported determination (20% at Descriptor level and at criterion level, respectively: 25%, 25%, 36%, 21% and 50%) and 10% used the same determination as last reported (35% at Descriptor level and at criterion level, respectively: 6%, 6%, 7%, 0% and 4%). This information is complementary to the information in section 3.2.

Recommendation for reporting: The many new GES determinations are caused by the poor reporting for these criteria in 2012. The new GES Decision considerably improved the quality for this reporting cycle; however, a commonly agreed GES determination, at least regionally, would reduce the importance of this field, as we would expect that most of the GES determinations would remain the same across the MSFD cycles.

3.3.2 Consistency and adequacy of reported GES spatial coverage

Table 17 provides some indicative GES determinations for the consistency and adequacy of reported GES spatial coverage (when relevant; regional, national, MRU) across the MS and within each region. This information is organised at the descriptor level on the top of Table 17 and then for each D6 criterion. The GES status was not defined for physical loss and

disturbance at regional level. At national level, few MS reported the maximum extent of thresholds.

Data: Member States 2018 reports (file in Annex: D6-Art.9; columns: GES components, GES description).

Table 17. Selection of GES determinations by MS, GES component and region.

GES comp.	MS	Reg	GES description
D6	PL	BAL	Regional threshold values have not been established yet. National values have been used instead. GES is determined on the basis of the modified indicator relevant for criterion D6C1 for WFD waters
D6	PL	BAL	The condition of the habitat type, including its biotic and abiotic structure and its functions (e.g. its typical species composition and their relative abundance, absence of particularly sensitive or fragile species or species providing a key function, size structure of species), is not adversely affected due to anthropogenic pressures. The threshold values are as follows: (a) in coastal waters, the values set in accordance with Directive 2000/60/EC; (b) beyond coastal waters, values consistent with those for coastal waters under Directive 2000/60/EC. Values were established through regional or subregional cooperation (HELCOM)
D6	PT	NEA	The evaluation of the Red List of Habitats in Europe considers that, for most of the habitats that occur in the North-East Atlantic, the existing data are insufficient and GES cannot be evaluated (Gubbay et al., 2016), although pressures, trends and conservation measures can be determined. According to the list, there are six threatened habitats, four classified as Vulnerable and two as Endangered. The habitats identified in the region [Azores] are distributed over a variety of ecological contexts and in a depth range ranging from 3 to 4 metres above the coast to depths of more than 4.000 m (supra-coastal to abyssal, Schmiing et al., 2015).
D6	CY	MED	The marine environment of Cyprus is considered to be in good environmental status by the year 2020 if the structure and function of the ecosystem are safeguarded and not adversely affected. Specifically, diversity and productivity are maintained, and any pressures do not hinder the ecosystem components to recover and/or retain their natural diversity, productivity and dynamic ecological processes. Given the available data, the determination of the GES of this descriptor relies heavily on criterion 6.2, which in turn is based on indicator 6.2.2. Indicator 6.2.2 is calculated using the various multimetric indices that have been developed in the framework of the Water-Framework Directive (2000/60/EC) for angiosperms, macroalgae and macrobenthos (and presently assigned a value of 0.9875).
D6C1	PL	BAL	GES is determined on the basis of the modified indicator relevant for criterion D6C1 for WFD waters.
D6C1	BE	NEA	Permanent changes to the seabed (km² or per cent relative to the natural size of the habitat) due to different human activities (including permanent changes to the seabed substrate or natural seabed morphology through physical restructuring, infrastructure development and loss of substrate through the extraction of seabed raw materials). Physical loss shall be understood as a permanent change to the seabed which has lasted or is expected to last for a period of two reporting cycles (12 years) or more. The assessment includes

GES comp.	MS	Reg	GES description
			quantification of the lost area compared to the total natural size of all benthic habitats in the area under assessment (e.g. due to the magnitude of the anthropogenic change).
D6C1	NL	NEA	No significant loss of the natural seabed compared to the situation in 2012 resulting from human activities.
D6C1	IE	NEA	Ireland has achieved Good Environmental Status within its maritime area under primary criterion D6C1 - spatial extent and distribution of physical loss (permanent change) of the natural seabed.
D6C1	CY	MED	The length of permanently altered coastline and estimated 100 m wide coastal water band have increased by 44% and 8% between 2012 and 2019. However, the actual fraction of affected coastline remains relatively low. A value of 0.75 is ascribed to this criterion.
D6C2	NL	NEA	No increase in time of the physical disturbance of the entire sea bed of the whole North Sea and the NCP.
D6C2	CY	MED	Percent change from 2012 to 2019. Based on the information compiled for this reporting, a conservative overall score of 0.75 is ascribed to this criterion.
D6C3	EE	BAL	GES under the criteria D6C3 is determined via the spatial area of physical disturbance to the habitat type sandbanks (code 1110), estuaries (code 1130), mudflats and sandflats (code 1140), large shallow inlets and bays (code 1160), reefs (code 1170). GES is achieved if the extent of the physically disturbed habitat is ≤ 10% of the total area of the habitat type.
D6C3	FI	BAL	The occurrence or quality of the habitat is not endangered by human activities that cause loss or disturbance to the seabed, and the magnitude of the disturbance must be commensurate with the ecological importance of the habitat and its threat. The integrity of the seabed is assessed for broad habitat types by marine area, with each broad habitat assessed according to the weakest status of the relevant indicators.
D6C3	ES	NEA	The extent of each benthic habitat type adversely affected by physical disturbances keeps negative or stable trends in a way that ensures their conservation.
D6C3	CY	MED	Assuming that the above indicators contribute equally to criterion D6C3, an average value of 0.8 can be ascribed to the criterion itself.
D6C3	ES	MED	The extent of each benthic habitat type adversely affected by physical disturbances keeps negative or stable trends in a way that ensures their conservation.
D6C3	MT	MED	The structure and functions of benthic habitats listed in Directive 92/43/EEC, in terms of species composition and relative abundance (as relevant), are in high/good status on the basis of biotic indices used under Directive 2000/60/EC for more than 75% of the area covered by the habitat.
D6C3	MT	MED	The structure and functions of MSFD predominant habitat types do not deviate from normal conditions and are in high/good status on the basis of biotic indices used under Directive 2000/60/EC, where established.
D6C4	EE	BAL	GES under the criteria D6C4 is determined via the spatial area of loss of the habitat type sandbanks (code 1110), estuaries (code 1130), mudflats and sandflats (code 1140), large shallow inlets and bays (code 1160) and reefs

GES comp.	MS	Reg	GES description
			(code 1170) resulting from anthropogenic loss of the natural seabed. GES is achieved if the habitat type loss resulting from anthropogenic loss of the natural seabed is ≤ 5% of the total area of the habitat type.
D6C4	FI	BAL	The distribution of benthic habitats is equivalent to their natural range and losses are identified only at local level. The state of the integrity of the seabed shall be evaluated for quality habitats by sea basin, so that each broad habitat is assessed according to the weakest status of the relevant indicators.
D6C4	BE	NEA	The extent of the loss of the habitat type as a result of anthropogenic loads is no greater than a determined part of the natural size of the habitat type in the area under assessment. Member States shall determine the maximum allowable extent of habitat loss as part of the total natural size of the habitat type, through cooperation at Union level, taking into account regional or sub-regional specificities.
D6C4	MT	MED	The area covered by benthic habitats listed in Directive 92/43/EEC is stable and is not smaller than the Favourable Reference Area values where established under Directive 92/43/EEC.
D6C5	FI	BAL	The value of BQI in the benthic communities in the open sea above the halocline (< 60m in depth) is 0,93 in the Gulf of Finland, 4,0 in the Northern Baltic Sea, 4,0 in the Åland Sea, 4,0 in the Bothnian Sea, 1,5 in the Quark and 1,5 in the Bay of Bothnia. Seabed integrity status shall be evaluated for broad habitat types per sea basin, so that each broad habitat is assessed according to the weakest status of the relevant indicators.
D6C5	FI	BAL	The concentration of dissolved oxygen on the seabed does not fall below the monthly average of 4 mg L-1. The seabed integrity status is assessed for broad habitat types by sea area, with each broad habitat being assessed according to the weakest status of the indicators that characterize them.
D6C5	LT	BAL	The species composition and relative abundance of macrofauna communities reach values indicating that the increase in nutrients and organic matter does not cause adverse effects. The indicator is assigned to criterion D5C8.
D6C5	PL	BAL	The condition of the habitat type, including its biotic and abiotic structure and its functions (e.g. its typical species composition and their relative abundance, absence of particularly sensitive or fragile species or species providing a key function, size structure of species), is not adversely affected due to anthropogenic pressures. The threshold values are as follows: (a) in coastal waters, the values set in accordance with Directive 2000/60/EC; (b) beyond coastal waters, values consistent with those for coastal waters under Directive 2000/60/EC. Values were established through regional or subregional cooperation (HELCOM).
D6C5	NL	NEA	The diversity of benthos does not show a decreasing trend in the assessed areas (OSPAR- assessment value).
D6C5	ES	MED	The extent of each habitat type in which benthic communities are kept within values to ensure their durability and performance is maintained or presented with increasing trends.

Findings: There are not enough comprehensive GES determinations to assess the sub(regional) consistency.

At the same level of GES determination, there is a large variety, for instance, some of them rephrased the Descriptor's definition while other addressed future implementation of monitoring or lack of regional agreements. Ten MS used D6 as GES component for a total of 20 determinations (four MS in BAL, four MS in NEA and four MS in MED). Of these, four MS repeated or rephased the GES Decision (BE, DK, HR, SI), two MS listed criteria (PT, LV) and five MS defined GES based on other Directives (i.e., Directive 2000/60/EC – WFD and HB), RSC (HELCOM) or agreements (DE, LV, IT, CY, PL, PT). Finally, two MS expressed that no GES determination was agreed yet (PL, PT). IE reported to have achieved GES for D6C1 and D6C4, however did not provide additional details for evaluating the status.

Fourteen MS reported on the GES criterion D6C3 with a total of 28 separate determinations (six MS in BAL with nine determinations, six MS in NEA with 10 determinations and six MS in MED with nine determinations). From the total, eight determinations mentioned some aspect of assessment methodology (e.g., integration rule: FI, indicators: LV, parameters: NL, and achieved values: CY), four MS repeated or rephased the descriptor or criteria definition (DE, FR, HR, FI), six MS expressed opinion based on other Directives, and nine MS reported that the component was not clear/relevant to them. One MS reported to have achieved GES without providing additional details.

Eleven MS reported D6C4 as a GES component with a total of 15 separate determinations (three MS in BAL with three determinations, five MS in NEA with six determinations, four MS in MED with five determinations and one MS in BLK with three determinations). Of these, 10 repeated or rephased the Descriptor definition, one MS reported based on other directives (HD), one MS mentioned some aspect of assessment methodology (e.g., integration: FI) and one definition was irrelevant to GES determination. IE stated that GES was achieved for D6C1 and D6C4 without providing details to evaluate the achievement.

Thirteen MS used D6C5 as a GES component with a total of 26 separate determinations (six MS in BAL with 12 determinations, six MS in NEA with seven determinations, three MS in MED with four determinations and one MS in BLK with one determination). Of these, five MS provided a relevant GES determination, five MS provided quantitative GES definitions, eight MS repeated or rephased the descriptor definition, and four MS expressed opinion based on other Directives (HD).

MT used other Directives for the GES description, HD for D6C1 and WFD for D6C4 for all criteria, while PL used WFD.

For the GES component D6C5, only a few target values were provided; however, it was not possible to assess their consistency by region. The few target values were:

- Habitat type quality indicator is above 0,6;
- Concentration of dissolved oxygen on the seabed does not fall below the monthly average of 4 mg L-1 (FI);
- Diversity of benthos does not show a decreasing trend in the assessed areas (OSPAR-assessment value) (NL); and
- Benthic communities' performance is maintained or presented with increasing trends (ES).

Gaps: Some MS grouped the GES determinations under a generic D6 GES component, which is neither quantitative nor informative. There are some references to other Directives, which did not provide a clear view of their use within the MSFD assessment.

BE provided GES determination for D6C4, but did not assess it.

Recommendation for reporting: The reporting guidance should provide a clear example of the information required to be included in the GES determination (e.g., elements to assess, parameter used and threshold value, integration type).

Recommendation for implementation: The many different GES determinations are caused by lack of a common approach to determine GES. A common agreed GES determination, at least regionally, will harmonise or better lead to a single well understood quantitative determination for the same component and habitat type. TG Seabed can recommend a common and harmonised approach to GES determination.

3.4 Justifications for delays for not using or setting EU/regional thresholds and for not reporting GES criteria

MS reported justifications for delays in determining GES and for not assessing particular criteria. This is an important source of information for gaps and for national or regional obstacles that can be prioritised and resolved with regional or EU collaboration.

Data: Member States 2018 reports (file in Annex: D6-Art.9; columns: Justification for of criterion, Justifications for delay in setting EU/regional requirements).

Table 18. Selection of justifications for non-use of criteria by MS.

CRITERIA	MS	JUSTIFICATION FOR NON-USE OF CRITERION
D6 /D1	DE	For the Habitats Directive habitat types 1110 sandbanks and 1170 reefs, the conservation status in the biogeographical region is used for the assessment of the benthic habitat (HD assessment 2013). There is no assessment of the individual MSFD criteria D6C3, D6C4 and D6C5.
	DE	For the Habitats Directive habitat types sandbanks (EU code 1110) and reefs (EU code 1170), which are rated as particularly protected benthic habitats, the conservation status of the HD assessment 2013 that applies to them in the continental biogeographical region is adopted. There is no assessment of the individual criteria D6C3, D6C4 and D6C5.
	PT	Discussed in the TG-Seabed Working Group in which Portugal participates.
	SI	The GES definition is set at the descriptor level according to the data currently available for condition assessment. One GES definition is set for D6, and another GES definition is set for criteria D6C4 and D6C5.
D6C1	EE	Criteria D6C1 assesses spatial distribution and area of physical loss of the natural seabed. According to Commission Decision 2017/848 and WG GES Guidance for assessments under Article 8 of the Marine Strategy Framework Directive criteria D6C1 does not have threshold value. It is used to assess other criteria. D6C1 is used to assess habitat loss per habitat under D6C4 (habitat extent (loss)), and to assess D7C1 (spatial extent and distribution of permanent alteration of hydrographical conditions).
	FI	This criterion does not have determination, because D6C1 is assessed under D6C4.
	HR	Assessment methodology not developed. Threshold values should be defined. No data available at the moment, GES not assessed.

CRITERIA	MS	JUSTIFICATION FOR NON-USE OF CRITERION
	LT	Seabed loss includes effects due to changes in the seabed substrate or morphology and the extraction of the seabed substrate, after which the natural properties of the seabed can only be restored by additional measures, as they are not restored in the course of natural processes. During the assessment period, no seabed infrastructure works were carried out to replace the seabed substrate. The criterion is not relevant.
	MT	D6C1 is considered to be covered through the following GES: "The area covered by benthic habitats listed in Directive 92/43/EEC is stable and is not smaller than the Favourable Reference Area values where established under Directive 92/43/EEC".
	SE	All Swedish determinations of GES are implemented in a regulation (HVMFS 2012:18) issued from the Swedish Agency for Marine and Water Management. From the update made in 2018 only criteria supported with at least one indicator are included in the regulation. Furthermore, only indicators underpinned with a regular monitoring programme are included. For this criterion, it is not relevant to have a definition of GES since the permanent changes to the seabed from different human activities shall be assessed to underpin the assessment under D6C4. In the assessment under article 8, an estimate of permanent changes is included. Concerning methods for estimation of permanent changes, we will follow and take active part in the work of TG Seafloor.
D6C2	EE	Criteria D6C2 assesses spatial distribution and area of physical disturbance to the seabed. According to Commission Decision 2017/848 and WG GES Guidance for Assessments Under Article 8 of the Marine Strategy Framework Directive criteria D6C2 does not have threshold value. It is used to assess other criteria. D6C2 is used to assess impacts on broad habitat types under D6C3.
	FI	This criterion does not have determination, because D6C2 is assessed under D6C5.
	HR	Assessment methodology not developed. Threshold values should be defined. No data available at the moment, GES not assessed.
	LT	The spatial extent and distribution of pressures associated with physical disturbance to the seabed within the predominant seabed habitat types were assessed. Impacts assessed: bottom trawling, excavation and burial, sanding to replenish beaches. The size of the physical disturbance area in square kilometers and the percentage of the physical disturbance area ratio were estimated. As both trawling and dumping and sand excavation take place in the same circalittoral habitats (although not spatially overlapping), it is proposed to establish a common indicator for all impacts. This is planned for the next phase of MSFD implementation.
	MT	D6C2 is considered to be covered through the following GES: "The structure and functions of benthic habitats listed in Directive 92/43/EEC, in terms of species composition and relative abundance (as relevant), are in high/good status on the basis of biotic indices used under Directive 2000/60/EC for more than 75% of the area covered by the habitat" and "The structure and functions of MSFD predominant habitat types do not deviate from normal conditions and are in high/good status on the basis of biotic indices used under Directive 2000/60/EC, where established".
D6C3	SE	All Swedish determinations of GES are implemented in a regulation (HVMFS 2012:18) issued from the Swedish Agency for Marine and Water Management. From the update made in 2018 only criteria supported with at least one indicator are included in the regulation. Furthermore, only indicators underpinned with a regular monitoring programme are included. For this criterion, it is not relevant to have a definition of GES since disturbance to the seabed from different human activities shall be assessed to underpin the assessment under D6C3. In the assessment under article 8, an

CRITERIA	MS	JUSTIFICATION FOR NON-USE OF CRITERION
		estimate of disturbance is included. Concerning methods for estimation of physical disturbance, we will follow and take active part in the work of TG Seafloor.
	FI	This criterion does not have determination, because D6C3 is assessed under D6C5.
	IE	This assessment did not cover criterion D6C3 (i.e. spatial extent of each habitat type which is adversely affected, through change in its biotic and abiotic structure and its functions [e.g. through changes in species composition and their relative abundance, absence of particularly sensitive or fragile species or species providing a key function, size structure of species], by physical disturbance). This was because agreed OSPAR Common Indicators or other coordinated indicators (e.g. at EU member state level) were not available for this criterion in OSPAR Region III or Region V, which are applicable to Ireland.
	LT	The spatial extent and distribution of pressures associated with physical disturbance to the seabed within the predominant seabed habitat types were assessed. Impacts assessed: bottom trawling, excavation and burial, sanding to replenish beaches. The size of the physical disturbance area in square kilometers and the percentage of the physical disturbance area ratio were estimated. As both trawling and dumping and sand excavation take place in the same circalittoral habitats (although not spatially overlapping), it is proposed to establish a common indicator for all impacts. This is planned for the next phase of MSFD implementation.
	RO	Lack of data
D6C4	LT	Due to the lack of data, a comprehensive assessment has not been carried out. Monitoring is planned for 2020-2023. Indicator in the HELCOM region under development. However, it is tentatively estimated that the extent of habitat type degradation due to anthropogenic pressures does not exceed a certain proportion of the natural size of the habitat type in the assessed area.
	PL	The criterion is not used in the assessment since there are no thresholds established and accepted on EU level for the extent of habitat loss.

Table 19. Selection of justifications for delay in setting EU/regional requirements by MS.

CRITERIA	MS	JUSTIFICATION FOR DELAY IN SETTING EU/REGIONAL REQUIREMENTS
D6 /D1	CY	Sufficient data is available for Descriptor to be assessed.
	DE	During the reporting period, regional cooperation focused on the development of methodological standards to assess the various aspects of stress and resilience on the basis of Commission Decision 2010/477/EU. The indicators vary in their level of maturity and require further development or testing. Commission Decision (EU) 2017/848 gives, for the first time, an explicit obligation on the part of the EU Member States to agree on evaluation elements, thresholds and integration rules in the framework of the EU MSFD CIS process and through regional or subregional cooperation. Germany is actively working with the countries bordering the Baltic Sea in the framework of the ongoing work programmes of the EU MSFD and the HELCOM bodies to establish where a coordinated definition is still missing.
	DE	Regional cooperation in the reporting period focused on the development of methodological standards for assessing the various pressure and condition aspects based on Commission Decision 2010/477/EU. In the regional or subregional cooperation, within the framework of OSPAR, it has mostly not been possible to agree

CRITERIA	MS	JUSTIFICATION FOR DELAY IN SETTING EU/REGIONAL REQUIREMENTS
		threshold values for these indicators in order to make (quantitative) statements about the state of the affected loads or the ecosystem components. Commission Decision (EU) 2017/848 results in an explicit obligation for the EU member states to agree assessment elements, threshold values and integration rules within the framework of the EU MSFD CIS process and through regional or subregional cooperation. Germany is actively working with the states bordering the North Sea, within the framework of the current work programmes of the EU MSFD CIS process and the OSPAR bodies, to establish them, where a coordinated definition is still missing.
	DK	Threshold values have not yet been set at EU, HELCOM or OSPAR levels.
D6C1	ES	It is necessary to establish the GES to meet the new criteria for D6. However, the definitions should be quite general as threshold and/or reference values have not yet been established.
	FR	Criterion D6C1 does not have a threshold required in Decision 2017/848/EU.
	HR	Assessment methodology not developed. Threshold values should be defined. No data available at the moment, GES not assessed.
	IE	There are currently no threshold values proposed for sea-floor integrity. The development of regional and sub-regional threshold values has not been undertaken at this time but is a work in progress at a European member state level.
	PL	Regional threshold values have not been established yet. National values have been used instead.
D6C2	ES	It is necessary to establish the GES to meet the new criteria for D6. However, the definitions should be quite general as threshold and/or reference values have not yet been established.
	FR	Criterion D6C2 does not have a threshold required in Decision 2017/848/EU.
	HR	Assessment methodology not developed. Threshold values should be defined. No data available at the moment, GES not assessed.
	IE	There are currently no threshold values proposed for sea-floor integrity. The development of regional and sub-regional threshold values has not been undertaken at this time but is a work in progress at a European member state level.
	LT	The spatial extent and distribution of pressures associated with physical disturbance to the seabed within the predominant seabed habitat types were assessed. Impacts assessed: bottom trawling, excavation and burial, sanding to replenish beaches. The size of the physical disturbance area in square kilometers and the percentage of the physical disturbance area ratio were estimated. As both trawling and dumping and sand excavation take place in the same circalittoral habitats (although not spatially overlapping), it is proposed to establish a common indicator for all impacts. This is planned for the next phase of MSFD implementation.
	PL	Regional threshold values have not been established yet. National values have been used instead.
D6C3	DK	At regional or sub-regional levels, thresholds must be established for when a physical disturbance will have a negative effect on each habitat type. Such thresholds have not yet been set by HELCOM or OSPAR.
	ES	It is necessary to establish the GES to meet the new criteria for D6. However, the definitions should be quite general as threshold and/or reference values have not yet been established.

CRITERIA	MS	JUSTIFICATION FOR DELAY IN SETTING EU/REGIONAL REQUIREMENTS
	FR	Criterion D6C3 requires the establishment, through cooperation between Member States at regional or sub-regional level, of threshold values for adverse effects of physical disturbance. Such thresholds have not yet been defined for the French marine sub-regions, but work in this direction is under way at the European level, particularly within the framework of the International Council for the Exploration of the Sea (ICES). The methodological standards associated with this criterion will be specified following further studies, as provided for in Ministerial Order 2019 on the definition of good environmental status of marine waters.
	LT	The spatial extent and distribution of pressures associated with physical disturbance to the seabed within the predominant seabed habitat types were assessed. Impacts assessed: bottom trawling, excavation and burial, sanding to replenish beaches. The size of the physical disturbance area in square kilometers and the percentage of the physical disturbance area ratio were estimated. As both trawling and dumping and sand excavation take place in the same circalittoral habitats (although not spatially overlapping), it is proposed to establish a common indicator for all impacts. This is planned for the next phase of MSFD implementation.
	PL	Regional threshold values have not been established yet. National values have been used instead.
	RO	No data for benthic habitats disturbance connected with physical disturbance
	SE	All Swedish determinations of GES are implemented in a regulation (HVMFS 2012:18) issued from the Swedish Agency for Marine and Water Management. From the update made in 2018 only criteria supported with at least one indicator are included in the regulation. Furthermore, only indicators underpinned with a regular monitoring programme are included. For this criterion, we only have an indicator for the North Sea giving a qualitative assessment. We do not have a definition of GES at criteria level yet. The reason is lack of enough and adequate information and knowledge on the benthic habitats. Sweden has an ongoing project on mapping the benthic habitats aiming at closing this knowledge gap. We are active in groups discussing this in HELCOM, OSPAR and ICES and we will be active in the TG Seafloor.
D6C4	CY	Not assessed
	DK	Thresholds are expected to be set for the extent of losses and negative effects per year, respectively habitat type under EU auspices. These thresholds have not yet been set.
	ES	It is necessary to establish the GES to meet the new criteria for D6. However, the definitions should be quite general as threshold and/or reference values have not yet been established.
	FI	GES D6C4 was not yet defined due to scientific knowledge gap, GES definition will be developed by 2024.
	FR	No indicators are available for the assessment of criterion D6C4. The methodological standards associated with criterion D6C4 will be further developed following additional studies, as provided for in the Ministerial Order 2019 on the definition of good environmental status of marine waters. The evaluation is available at: https://doi.org/10.13155/60192
	IE	There are currently no threshold values proposed for sea-floor integrity. The development of regional and sub-regional threshold values has not been undertaken at this time but is a work in progress at a European member state level.

CRITERIA	MS	JUSTIFICATION FOR DELAY IN SETTING EU/REGIONAL REQUIREMENTS
	LT	Due to the lack of data, a comprehensive assessment has not been carried out. Monitoring is planned for 2020-2023. Indicator in the HELCOM region under development. However, it is tentatively estimated that the extent of habitat type degradation due to anthropogenic pressures does not exceed a certain proportion of the natural size of the habitat type in the assessed area.
	SE	All Swedish determinations of GES are implemented in a regulation (HVMFS 2012:18) issued from the Swedish Agency for Marine and Water Management. From the update made in 2018 only criteria supported with at least one indicator are included in the regulation. Furthermore, only indicators underpinned with a regular monitoring programme are included. We on not have any indictors under this criterion. The reason is lack of enough and adequate information and knowledge on the benthic habitats and that threshold values shall be established at union level. We will include thresholds when they are established at union level. We are active in groups discussing this in Helcom, Ospar and ICES and we will be active in, and look forward to, the work on this in CIS-MSDF.
D6C5	CY	Not assessed
	DK	Thresholds for negative effects on the state of each habitat type are also expected to be set in relation to the influence of other pressure factors (descriptors) as well as the extent of negative effects per year on habitat type under EU auspices. These thresholds have not yet been set.
	ES	It is necessary to establish the GES to meet the new criteria for D6. However, the definitions should be quite general as threshold and/or reference values have not yet been established.
	FR	The assessment of the D6C5 criterion is informed by the calculation of the BenthoVal indicator (Labrune et al., in preparation) selected at the national level. Due to the absence of a defined threshold, criterion D6C5 is not assessed. It should be noted that for this assessment, the BenthoVal indicator could not be calculated for rocky habitats, despite the acquisition of data, because the applicability of BenthoVal to these types of habitats requires further testing. The methodological standards associated with criterion D6C5 will be specified following further studies, as provided for in Ministerial Order 2019 on the definition of good ecological status of marine waters. The evaluation is available at: https://doi.org/10.13155/60192
	IE	There are currently no threshold values proposed for sea-floor integrity. The development of regional and sub-regional threshold values has not been undertaken at this time but is a work in progress at a European member state level.
	LT	Not all habitat types are subject to macrozoobenthos monitoring, and the existing network of stations does not allow for spatial assessment of habitat status everywhere. Due to the lack of data, the assessment is not comprehensive according to the criteria.
	SE	All Swedish determinations of GES are implemented in a regulation (HVMFS 2012:18) issued from the Swedish Agency for Marine and Water Management. From the update made in 2018 only criteria supported with at least one indicator are included in the regulation. Furthermore, only indicators underpinned with a regular monitoring programme are included. Under this criterion, we have used proxies from other descriptors (D5) for the assessment, which includes assessments under Directive 2000/60/EC. In addition, we have used assessment from the Directive 92/43/EEC, but the assessment is not complete in relation to the broad habitat types. The reason is

CRITERIA	MS	JUSTIFICATION FOR DELAY IN SETTING EU/REGIONAL REQUIREMENTS
		lack of enough and adequate information and knowledge on the benthic habitats and that threshold values shall be established at union level. We will include thresholds when they are established at union level. We are active in groups discussing this in HELCOM, OSPAR and ICES and we will be active in the work on this in CIS-MSFD.

Findings:

Justification for not determining GES

Under the general GES component D6, three MS (DE in BAL and NEA, SI in MED, PT in NEA) reported justifications for not providing a GES determination because of missing EU/regional agreement.

In addition, 29 justifications for non-use of specific criteria were provided by 12 MS. For GES component D6C1, six MS (four in BAL, two in MED, and one in NEA) reported that agreed threshold values are missing. Two MS (one in BAL, one in MED) reported that the criterion was already assessed under D6C4. Three MS reported that the criterion was already covered by HD or that the GES determination was not relevant.

For GES component D6C2, six MS (four MS in BAL, two MS in MED, and one MS in NEA) provided justifications. Most of them reported that agreed threshold values are missing. Three MS reported that the criterion was assessed under criteria D6C3/D6C5 and one MS reported that the criterion was already covered by HD and WFD. Finally, on MS stated that a GES determination was not relevant for D6C2.

Four MS provided a justification for the GES component D6C3 (three MS in BAL, one MS in NEA, and one MS in BLK). They reported that agreed threshold values are missing, that the criterion was assessed under D6C5 and there was a lack of data. Two MS stated that a GES determination was not relevant for D6C3.

For GES component D6C4, two MS in BAL reported lack of data, while threshold values were not established yet. And for the GES component D6C5 no justifications for not determining GES were reported.

Justification for delay in setting EU/regional requirements by MS

The D6 criteria are all primary, so there are no cases of prioritization of criteria, as for other descriptors. The MS reported 69 justifications for delays, of which five were under the GES component D6 stating either that they were working with RSC and EU on an agreed determination (two MS in BAL, two MS in NEA) or that there was not sufficient available data (one in MED).

For the GES component D6C1, the MS reported that regional thresholds are missing, no threshold values were required by the Decision and no GES determination or information on the assessment methodology was available.

For the GES component D6C2, six MS provided 11 justifications, including missing regional threshold values and that no value is required by the Decision for D6C2.

For GES component D6C3, seven MS provided 17 justifications stating that a regional agreement at RSC/EU level is missing.

For GES component D6C4, eight MS provided 15 justifications, and for D6C5, seven MS provided 14 justifications, where they reported that the criterion was not assessed or that there are no threshold values established yet.

Once again in most cases missing regional, EU or RCS agreement on methodology (indicators, threshold values) was the main reason given by the MS.

DE reported to not having assessed for D6C3, D6C4 and D6C5 without providing a proper justification, however in Art.8 for D6C3 specifies that "*The HELCOM-developed indicator (Cumulative Impact on benthic biotopes) for the criteria D6C1, D6C2, D6C3 could not be used due to a lack of regionally agreed thresholds. In addition, the Baltic Sea Impact Index (BSII) developed descriptively by HELCOM and used descriptively for criterion D6C3 takes into account both the physical loss (criterion D6C1) and the physical disturbance (criterion D6C2), as provided for in Commission Decision 2017/848 / EU*."

Gaps: The messages derived from this reporting field are clear and mostly related with gaps in data, in methods, in thresholds and in agreed regional or EU methodology.

Recommendation for reporting: In some cases, the MS provided justifications for delays, although a GES determination was also reported, because there was no assessment of the criteria for Art. 8 or because a quantitative threshold value is missing. In these cases, the reporting guidance could clarify that MS still can report qualitative determinations. Furthermore, some justifications are too wordy, and it was not easy to interpret and analyse. Possible grouping of causes led to misreporting, so based on the current experience, if organised in a drop-down menu it could facilitate both the reporting and the analysis.

Recommendation for implementation: This field is very useful as it provides gaps and drawbacks that did not allow MS to report GES determination or assess a particular criterion. Once again, the MS reported similar information in an inconsistent way, which was not easy to be analysed. The recommendation for reporting could overcome this problem and amend the reporting guidance accordingly. TG Seabed can contribute to tackle the gaps and drawbacks reported by the MS regarding GES determination and criteria usage. Moreover, the TG ensures the regional or EU coordination across the MS.

4 Results of Art.10 analysis for D6 benthic habitats

4.1 Key results and recommendations

- 714 targets were reported for D6 under Art. 10 by 20 MS. The targets reported varying from 1 to 195 per MS.
- The analysis revealed data gaps, which are evident by the inability of the MS to provide measurable targets for all criteria.
- There is a complete lack of measurable targets with associated target values for the reported parameters. Thus, the targets are unable to measure the progress toward GES.
- There is a lack of consistency in the target setting and lack of any regional coordination.
- The reported elements do not follow a consistent pattern across the MS and similarly with the fluctuating number of targets reported per MS the elements are signed inconsistently across MS.
- Some target values are set, once again in a way which is not regionally coordinated and do not allow comparison across MS. All target values correspond to indicators or criteria that were reported for Art. 8 and do not refer to measurable targets able to estimate the progress toward GES achievement.
- A guidance document focused on MSFD Art. 10 target is required, following a similar organisation to the Art. 8 guidance, to ensure a common understanding across the MS and enforce a regional coordination for setting regional targets. This would facilitate the monitoring and evaluation of the GES achievement for D6/D1 – benthic habitats, at the proper ecological scale and at regional or subregional level.

4.2 Spatial consistency of environmental targets at EU and regional level

Overall, 714 targets were reported by 20 MS: 105 for BAL, 455 for NEA, 139 for MED and 15 for BLK. MS reported from one (CY) up to 195 targets (ES combined targets for MED and NEA) (Figure 6).

Data: Member States 2018 reports (file in Annex: D6-Art.10; columns: Description target for Figure 6, Target status for Table 20 and Figure 7, and update type Tables 21-26 and Figures 7-13).

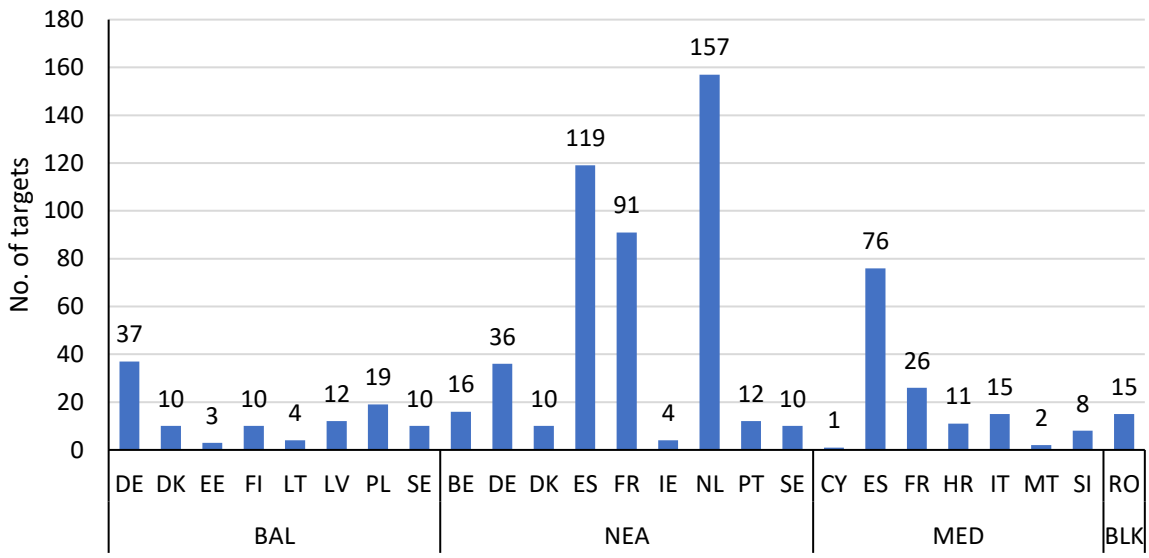

Figure 6. Number of targets reported for D6 by the 20 MS per marine region.

Table 20. Proportions of the target status reported per marine region.

Region	Target achieved	Target not yet achieved	Status not reported	Total number of targets
BAL	13	53	39	69
NEA	73	202	180	374
MED	11	38	90	103
BLK	14	1		15

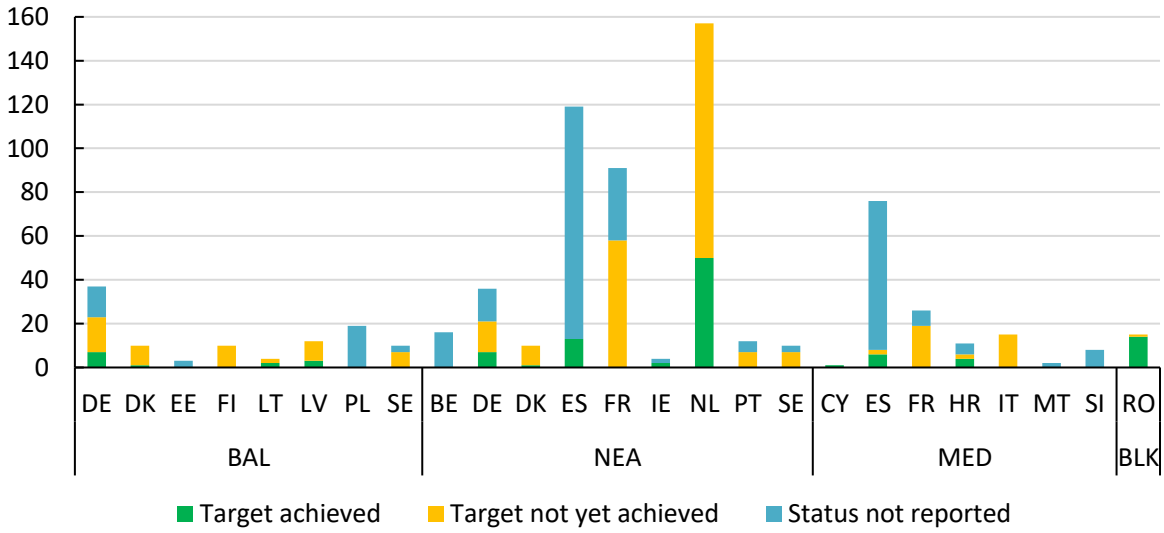

Figure 7. Target status reported by MS and marine regions.

Findings: Overall, 714 targets were reported; 105 for BAL, 455 for NEA, 139 for MED and 15 for BLK.

Of the 27 GES components (descriptor level and combination of criteria) reported, five addressed only one criterion, nine multiple criteria, one component referred to one descriptor and 12 to multiple descriptors. Tables 21-26 reports the number of targets by GES component.

From the total number of reported targets about 16% were achieved (12% of targets reported in BAL, 16% in NEA, 9.7% in MED, and 93.3% in BLK, respectively) and 40% not achieved (51% in BAL, 44.3% in NEA and 15.9% in MED and 6.6% in BLK). For the remaining targets (44%) the status was not reported (Table 20 and Figure 7). The main reasons for not reporting target status include: i) targets were no longer relevant, ii) no need to assess environmental target, but only the operational targets, and iii) the target was merged with another one or changed to comply with the new GES Decision.

The analysis of the targets by the level of their GES component showed that 476 targets were at Descriptor level, of which 434 refer only to D6 and the remainder addressed in addition one or more criteria, while 6 addressed more GES components (FR). Of the targets addressing criteria, 89 were addressing only one criterion and 149 were addressing several criteria.

Table 21 presents all the targets at Descriptor level. Table 22-26 reports all targets at criterion level, for each criterion. Figure 8 presents the targets' update type and status at Descriptor level, while Figure 9-13 present the same information at criterion level.

Table 21. Number of targets at D6 or D1/D6 Descriptor level by MS, update type and target status. The targets highlighted in grey were also reported for other Descriptors.

MS	Subregion	No. target	Update type	Target status
BE	ANS	4	Same as 2012 definition	(blank)
BE	ANS	6	Target no longer needed	(blank)
BE	ANS	4	Modified from 2012 definition	None
BE	ANS	1	New target	None
CY	MAL	1	Same as 2012 definition	Target achieved
DE	ANS BAL	14	Same as 2012 definition	Target achieved
DE	BAL	2	Modified from 2012 definition	Target not yet achieved
DE	ANS BAL	28	Same as 2012 definition	Target not yet achieved
DE	ANS BAL	29	Same as 2012 definition	(blank)
DK	ANS BAL	2	New target	Target achieved
DK	ANS BAL	2	New target	Target not yet achieved
EE	BAL	1	New target	None
ES	ABI AMA MWE	19	Modified from 2012 definition	Target achieved
ES	MWE	2	Modified from 2012 definition	Target not yet achieved
ES	ABI AMA MWE	69	Modified from 2012 definition	(blank)
ES	ABI AMA MWE	10	Target no longer needed	(blank)
ES	ABI AMA MWE	65	New target	None
FI	BAL	5	New target	Target not yet achieved

MS	Subregion	No. target	Update type	Target status
FI	BAL	4	Same as 2012 definition	Target not yet achieved
FR	ABI ACS ANS MWE	77	New target	Target not yet achieved
FR	ABI ACS ANS MWE	4	New target	None
FR	ABI ACS ANS	30	Target no longer needed	None
FR	MWE	6	Target no longer needed	None
HR	MAD	1	Target no longer needed	(blank)
IT	MAD MIC MWE	3	Same as 2012 definition	Target not yet achieved
LT	BAL	2	Same as 2012 definition	Target achieved
LT	BAL	2	Same as 2012 definition	Target not yet achieved
LV	BAL	3	New target	Target achieved
LV	BAL	9	New target	Target not yet achieved
PL	BAL	19	New target	(blank)
PT	AMA	3	Modified from 2012 definition	Target not yet achieved
PT	ABI	2	New target	Target not yet achieved
PT	AMA	2	Same as 2012 definition	Target not yet achieved
PT	AMA	1	Modified from 2012 definition	(blank)
PT	AMA	1	Same as 2012 definition	(blank)
PT	AMA	2	New target	None
PT	AMA	1	Same as 2012 definition	None
SE	ANS BAL	2	Same as 2012 definition	Target not yet achieved
SE	ANS BAL	2	Modified from 2012 definition	(blank)
SI	MAD	1	New target	None
SI	MAD	5	Target no longer needed	None

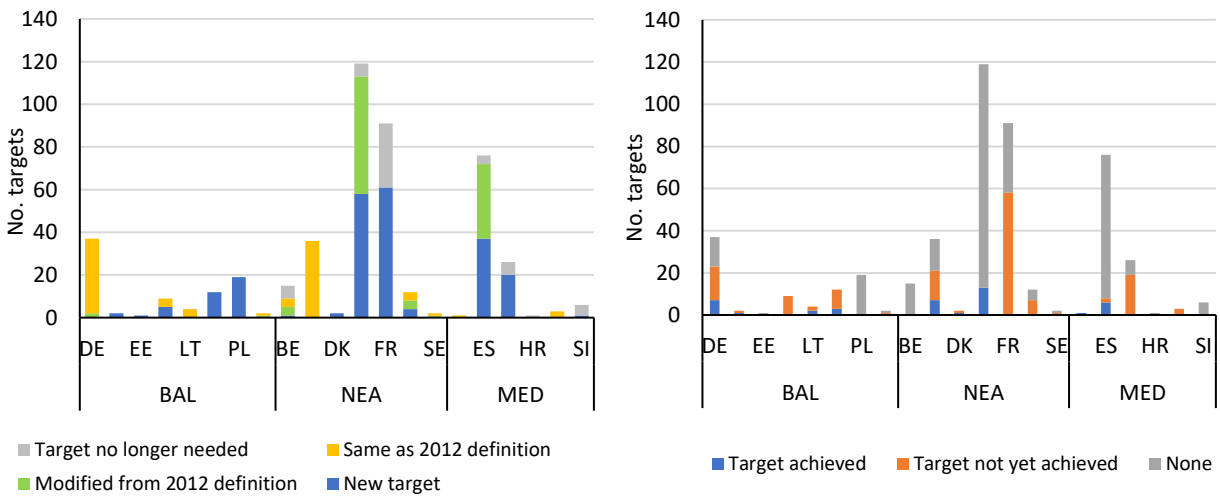

Figure 8. Update type (left) and target status (right) of the targets at Descriptor D1/D6 and GES component levels reported by each MS by region.

The highlighted row in Table 21 refers to the targets that address the GES component D1/D6. FR reported 6 targets at multiple Descriptor level; all these targets were reported as no longer needed; possibly with the intention to revise their targets.

From the 476 total number of targets at descriptor level, 225 (47,3%) were new targets, 103 (21.6%) modified targets, 93 (19,5%) same as 2012 definition and 58 (12.2%) targets no longer needed. From the total number of targets, 41 (8.6%) targets were reported as achieved, 143 (30.0%) targets were reported as not achieved and 292 (61.3%) had no reported status. Almost 50% of all targets were reported by ES and NL.

Table 22. Summary of targets at D6C1 level by MS, update type and target status.

MS	Subregion	No. target	Update type	Target status
DK	ANS	4	Target not yet achieved	New target
DK	BAL	4	Target not yet achieved	New target
EE	BAL	1	None	Modified from 2012 definition
EE	BAL	1	None	New target
HR	MAD	1	None	New target
IE	ACS	1	Target achieved	New target
IT	MAD	2	Target not yet achieved	Modified from 2012 definition
IT	MIC	2	Target not yet achieved	Modified from 2012 definition
IT	MWE	2	Target not yet achieved	Modified from 2012 definition
IT	MAD	1	Target not yet achieved	Same as 2012 definition
IT	MIC	1	Target not yet achieved	Same as 2012 definition
IT	MWE	1	Target not yet achieved	Same as 2012 definition
NL	ANS	13	Target not yet achieved	Same as 2012 definition
PL	BAL	4	(blank)	New target
SE	ANS	1	(blank)	Modified from 2012 definition
SE	BAL	1	(blank)	Modified from 2012 definition

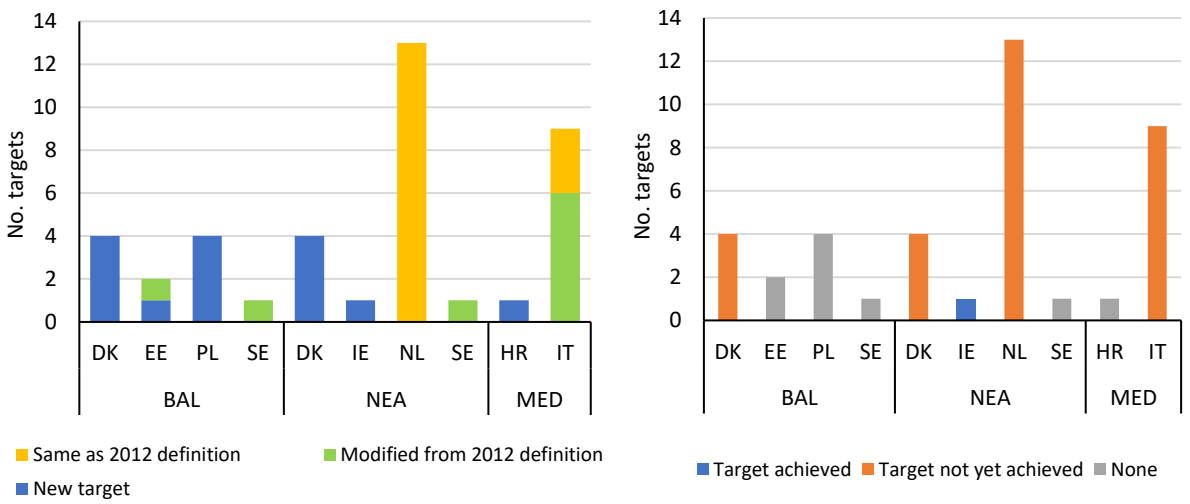

Figure 9. Update type (left) and target status (right) of the targets at criteria D6C1 level reported by each MS by region.

From the 40 targets reported at D6C1 level, 15 were new, nine were modified from 2012 definition and 16 were the same as 2012. Only one target was reported as achieved (IE), 30 as not achieved and nine had no reported status.

Table 23. Summary of targets at D6C2 level by MS, update type and target status.

MS	Subregion	No. target	Update type	Target status
DK	ANS BAL	8	Target not yet achieved	New target
EE	BAL	1	None	New target
HR	MAD	1	None	New target
IE	ACS	1	(blank)	New target
IT	MAD MIC MWE	6	Target not yet achieved	Modified from 2012 definition
IT	MAD MIC MWE	3	Target not yet achieved	Same as 2012 definition
MT	MIC	2	(blank)	Modified from 2012 definition
NL	ANS	49	Target achieved	Same as 2012 definition
NL	ANS	50	Target not yet achieved	Same as 2012 definition
PL	BAL	3	(blank)	New target
SE	ANS BAL	2	(blank)	Modified from 2012 definition

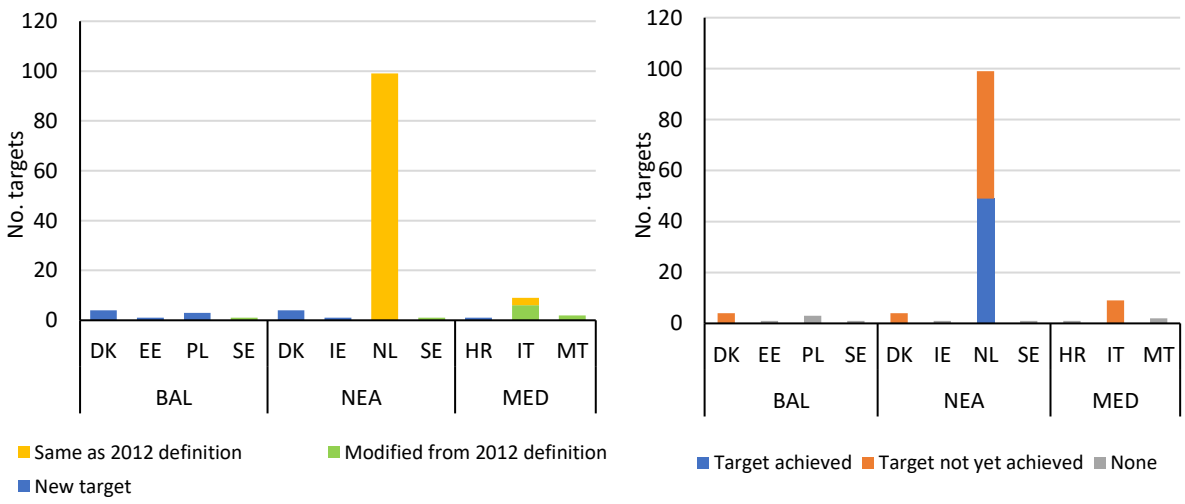

Figure 10. Update type (left) and target status (right) of the targets at criteria D6C2 level reported by each MS by region.

From the 126 targets reported at D6C2 level, 14 were new targets, 10 were modified from 2012 definition and 102 were the same as 2012. Of these, 49 targets were reported as achieved, 67 as not yet achieved and for 10 no status was reported.

Table 24. Summary of targets at D6C3 level by MS, update type and target status.

MS	Subregion	No. target	Update type	Target status
BE	ANS	3	(blank)	Same as 2012 definition
DK	ANS BAL	10	Target not yet achieved	New target
HR	MAD	1	Target achieved	Modified from 2012 definition
MT	MIC	2	(blank)	Modified from 2012 definition
NL	ANS	26	Target achieved	Same as 2012 definition
NL	ANS	95	Target not yet achieved	Same as 2012 definition
PL	BAL	3	(blank)	New target
SE	ANS	1	(blank)	Modified from 2012 definition

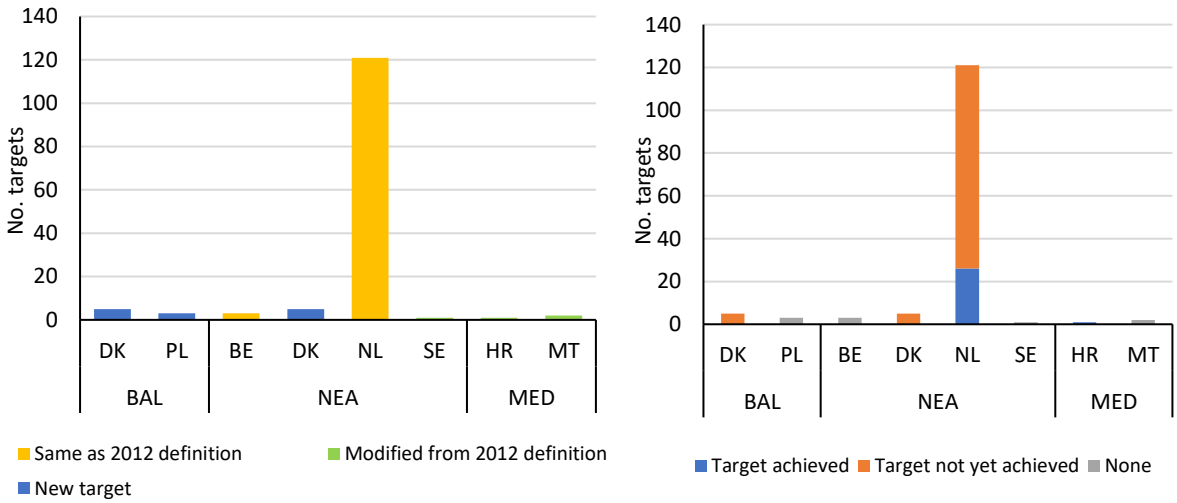

Figure 11. Update type (left) and target status (right) of the targets at criteria D6C3 level reported by each MS by region.

Of a total of 141 targets at D6C3 level, 13 were new targets, four were modified from 2012 definition and 124 were the same as 2012. Of these, 27 targets were reported as achieved, 105 as not yet achieved and for nine no status was reported.

Table 25. Summary of targets at D6C4 level by MS, update type and target status.

MS	Subregion	No. target	Update type	Target status
BE	ANS	1	(blank)	Same as 2012 definition
DK	ANS BAL	12	Target not yet achieved	New target
HR	MAD	1	Target achieved	Same as 2012 definition
HR	MAD	2	Target not yet achieved	Target no longer needed
HR	MAD	1	(blank)	Target no longer needed
IE	ACS	1	Target achieved	New target
IT	MAD MIC MWE	3	Target not yet achieved	Modified from 2012 definition
NL	ANS	25	Target achieved	Same as 2012 definition

MS	Subregion	No. target	Update type	Target status
NL	ANS	38	Target not yet achieved	Same as 2012 definition
PL	BAL	1	(blank)	New target
RO	BLK	8	Target achieved	New target
SE	ANS	1	Target not yet achieved	Same as 2012 definition
SI	MAD	1	None	New target

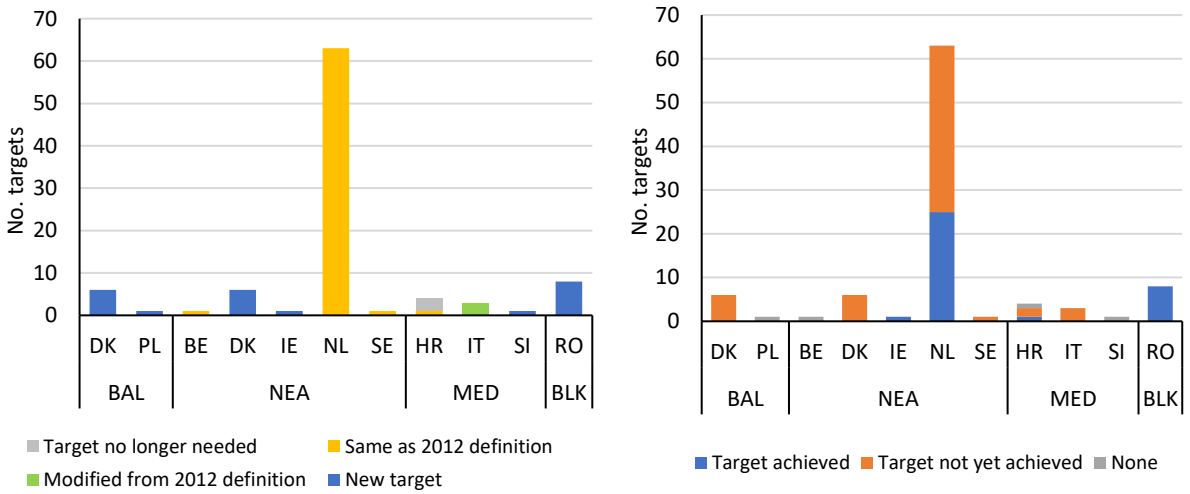

Figure 12. Update type (left) and target status (right) of the targets at criteria D6C4 level reported by each MS by region.

Of a total of 126 targets at D6C4 level, 23 were new targets, three were modified from 2012 definition, 66 were the same as 2012, and three were no longer needed. Of these, 35 targets were reported as achieved, 56 as not yet achieved and for four no status was reported.

Table 26. Summary of targets at D6C5 level by MS, update type and target status.

MS	Subregion	No. target	Update type	Target status
BE	ANS	2	(blank)	Same as 2012 definition
BE	ANS	6	(blank)	Target no longer needed
BE	ANS	4	None	Modified from 2012 definition
BE	ANS	1	None	New target
DK	ANS BAL	8	Target not yet achieved	New target
FI	BAL	1	Target not yet achieved	Same as 2012 definition
HR	MAD	1	Target achieved	Same as 2012 definition
HR	MAD	1	Target achieved	Target no longer needed
HR	MAD	1	(blank)	Same as 2012 definition
IE	ACS	1	(blank)	New target
IT	MAD MIC MWE	1	Target not yet achieved	Modified from 2012 definition
MT	MIC	2	(blank)	Modified from 2012 definition

MS	Subregion	No. target	Update type	Target status
NL	ANS	26	Target achieved	Same as 2012 definition
NL	ANS	82	Target not yet achieved	Same as 2012 definition
PL	BAL	7	(blank)	New target
RO	BLK	6	Target achieved	Modified from 2012 definition
RO	BLK	1	Target not yet achieved	Modified from 2012 definition
SE	ANS BAL	6	Target not yet achieved	Modified from 2012 definition
SE	ANS BAL	2	(blank)	Modified from 2012 definition
SE	ANS BAL	1	(blank)	Same as 2012 definition
SI	MAD	1	None	New target

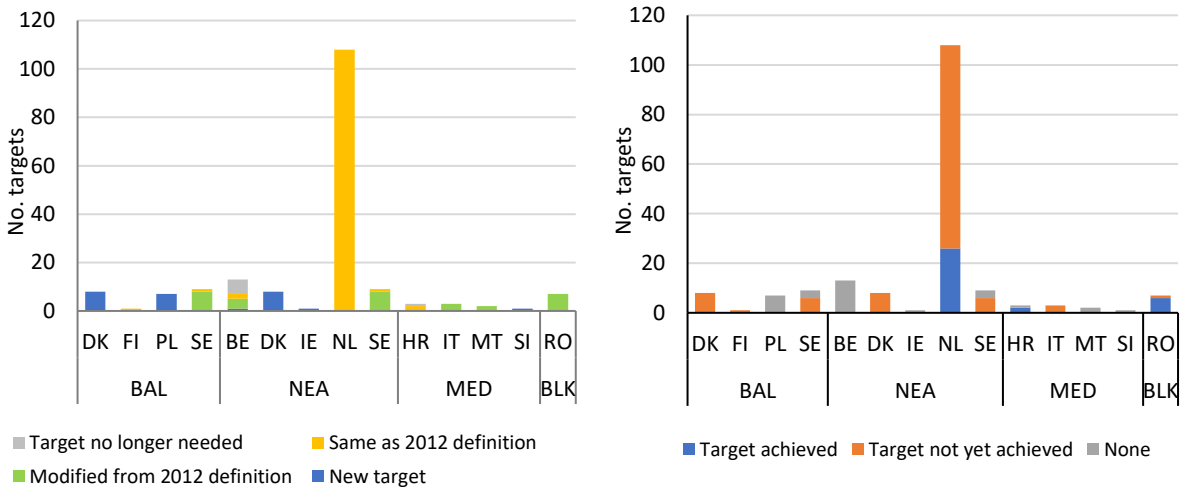

Figure 13. Update type (left) and target status (right) of the targets at criteria D6C5 level reported by each MS by region.

Of a total of 126 targets at D6C5 level, 26 were new targets, 32 were modified from 2012 definition and 115 were the same as 2012, and seven were no longer needed. Of these, 34 targets were reported as achieved, 115 as not yet achieved and for 31 no status was reported.

An indicative selection of targets by GES component is reported in the Tables 27 - 31 to show the variety of the reported information under the same GES component and the inconsistency across the MS.

Data: Member States 2018 reports (file D6-Art.10; columns: Description target, update type, target status).

Table 27. Description of targets at D6C1 level by MS, update type and target status.

MS	MSFD subregion	Description target	Update type	Target status
DK	ANS BAL	Through regional work and the work in the EU, a better understanding of the impacts on the seafloor in relation to loss, disturbance and adverse effects is achieved. (Operational target)	New target	Target not yet achieved
DK	ANS BAL	When threshold values for losses, disturbances and adverse effects are established through cooperation at regional and Union level, the Ministry of Environment and Food will initiate a project to form the basis for establishing targets in accordance with the thresholds and good environmental status. (Operational target)	New target	Target not yet achieved
EE	BAL	BALEE-T30 - coastal impact index is showing improvement trend towards achieving good environmental status and/or is not declining in the assessment units where good environmental status has already been achieved.	New target	None
HR	MAD	Spatial extent and distribution of physical loss of the natural seabed does not increase to an extent that endangers the functioning of the ecosystem.	New target	None
IT	MAD MIC MWE	Vessels operating with fishing gear which interact with the seabed in active is equipped with tools for recording and reporting position of vessels themselves, in particular those equipped with hydraulic dredges and bottom trawls with LOA <Â 15 m.	Same as 2012 definition	Target not yet achieved
NL	ANS	No increase in time of physical disturbance from fishing activities of the total seabed of the NCP and of the habitats described under the MSFD.	Same as 2012 definition	Target not yet achieved

Table 28. Description of targets at D6C2 level by MS, update type and target status.

MS	MSFD subregion	Description target	Update type	Target status
EE	BAL	BALEE-T30 - coastal impact index is showing improvement trend towards achieving good environmental status and/or is not declining in the assessment units where good environmental status has already been achieved.	New target	None
HR	MAD	Spatial extent and distribution of physical disturbance pressures on the seabed are in an extent that enables sustainable use of ecosystem.	New target	None

MS	MSFD subregion	Description target	Update type	Target status
IE	ACS	The spatial extent and distribution of physical disturbance pressures on the seabed is at a level that ensures that the structure and functions of the ecosystems, and benthic ecosystems, in particular, are not adversely affected	New target	(blank)
IT	MAD MIC MWE	At least 10% of the area with substrates exploited by fishing activities that interact with the seabed in an active way is protected.	Modified from 2012 definition	Target not yet achieved
MT	MIC	The intensity and distribution of otter bottom trawling are stable and do not increase significantly through time.	Modified from 2012 definition	(blank)
NL	ANS	Restoration of rest for marine mammals and birds by reducing fishing in the Vlakte van de Raan and the North Sea Coastal Zone (under the VIBEG agreement).	Same as 2012 definition	Target achieved Target not yet achieved

Table 29. Description of targets at D6C3 level by MS, update type and target status.

MS	MSFD subregion	Description target	Update type	Target status
BE	ANS	Positive trend in sea floor surface area permanently devoid of bottom-contacting fishing gear disturbance within each of the benthic habitat types (= pressure indicator), as to allow a natural development of the benthic fauna and flora and as to minimise artificial fragmentation of the seafloor (= desired state).	Same as 2012 definition	(blank)
BE	ANS	The spatial extent and distribution of the EUNIS level 3 habitats (sandy mud to mud, muddy sands to sands and coarse grained sediments), as well as that of gravel beds fluctuate - relative to the reference state as described in Initial Assessment - within a margin limited to the accuracy of the current distribution maps	Same as 2012 definition	(blank)
HR	MAD	Intertidal hard bottom and rocks: a) Good ecological status of benthic communities on hard bottoms is not decreasing. b) Distribution of biocenosis is maintained. Biocenosis of sublittoral algae: a) Good ecological status of benthic communities on hard bottoms	Modified from 2012 definition	Target achieved

MS	MSFD subregion	Description target	Update type	Target status
		is not decreasing. b) Distribution of biocenosis is maintained.		
NL	ANS	10-15% of the surface of the Dutch part of the North Sea is not significantly affected by human activities.	Same as 2012 definition	Target not yet achieved
NL	ANS	Further development and testing of regional assessment methods (OSPAR) that can be used in the future for the assessment of benthic and pelagic habitats.	Same as 2012 definition	Target not yet achieved
PL	BAL	Limiting cumulative pressures on habitats.	New target	(blank)

Table 30. Description of targets at D6C4 level by MS, update type and target status.

MS	MSFD subregion	Description target	Update type	Target status
HR	MAD	a) Coastline colonized by Cystoseira amantacea is stable or increasing and not smaller that the baseline value. b) The upper and lower limits of Posidonia meadows are not in regression.	Target no longer needed	(blank)
HR	MAD	a) Range of photophilic algal community is stable or increasing according to all indicators monitored and not below then Favourable Reference Range (according to Habitat Directive). b) Range of Posidonia meadow is stable or increasing and is not less than the FRR value (Favourable Reference Range for HD habitats). c) Distributional range of Coralligenous assemblages in Croatia is stable or increasing and not smaller than the FRR value (Favourable Reference Range for HD habitats).	Target no longer needed	Target not yet achieved
HR	MAD	Posidonia oceanica meadow: a) The area of distribution of P. oceanica meadow is not decreasing. b) Ecological quality of P. oceanica meadow is not decreasing.	Same as 2012 definition	Target achieved
IT	MAD MIC MWE	Increase in the number of marine habitats listed in the Habitats Directive and referred to in the SPA/BD Protocol of the Barcelona Convention which maintains or achieves a favourable conservation status	Modified from 2012 definition	Target not yet achieved
RO	BLK	The extent of loss of each habitat type, resulting from anthropogenic pressures, does not exceed 5% of the natural extent of the habitat type in the marine unit.	New target	Target achieved

MS	MSFD subregion	Description target	Update type	Target status
SI	MAD	Prevention of additional loss of benthic habitat types in the mediolittoral and infralittoral. Maintaining the good status of benthic habitats in the circalittoral.	New target	None

Table 31. Description of targets at D6C5 level by MS, update type and target status.

MS	MSFD subregion	Description target	Update type	Target status
BE	ANS	Decreasing relative frequency of occurrence of damaged Asterias rubens (2+ cm arm length) and tube clusters of Pomatoceros triqueter, indicative for physical disturbance of the bottom (= pressure indicator), as to enhance natural development of the gravel bed ecosystem (= desired state).	Target no longer needed	(blank)
BE	ANS	The bioturbation potential (BPc), an indicator for the functioning of the benthic ecosystem, has a minimum value of 0.60 for the Abra alba habitat type in the autumn (as determined by BEQI procedure).	Modified from 2012 definition	None
FI	BAL	Response capacity for oil and chemical accidents is secured.	Same as 2012 definition	Target not yet achieved
PL	BAL	Reducing the pressures associated with the introduction of biogenic, synthetic, non-synthetic substances, organic matter into the environment as well as related with extraction and compaction of the substrate causing secondary abrasion and siltation in order to restore close to the natural structure of macrophyte and angiosperm plant communities. In terms of introduction of substances, the targets are consistent with those for D5. In terms of impact on the seabed, the targets are consistent with those for the criteria D6C1, D6C2, D6C3 and D7C1 and D7C2.	New target	(blank)
RO	BLK	75% of M-AMBI*(n) values > 0.61 in infralittoral sand and circalittoral mud	Modified from 2012 definition	Target achieved
SE	ANS BAL	C.1 The marine environment shall be free of newly-released or newly-translocated non-indigenous species and strains, and non-indigenous species otherwise introduced through human activities, which may	Modified from 2012 definition	(blank)

MS	MSFD subregion	Description target	Update type	Target status
		adverse effects on genetic or biological diversity or the ecosystem functioning.		

Findings: The analysis of the targets highlights the lack of a common agreement on how to consistently set environmental targets that address particular pressures preventing GES achievement. Moreover, the targets are not expressed quantitatively as an amount of pressure reduction that is to be achieved in order to get the pressure down to GES-compatible levels or to achieve GES (reductions in impacts). The quantification should be centred on the gap between current state and GES (i.e., how much reduction is needed) and not another expression of Art. 9 or a repeat of the Art. 8 assessment.

Overall, targets are not quantifying the gap between the current state and GES. Many MS provided generic targets, some rephrasing the descriptor definition (CY, HR, IE, NL, PL and SE) or expressing the need to establish threshold values at EU level (DK). Several MS provided targets using trends (BE, EE, HR, MT) or providing indicators or spatial extents (NL, RO). DK reported a target related to a monitoring programme (NOVANA).

For D6C4 and D6C5, several MS mentioned other directives, such as the HD (DK, HR, IT, SE), WFD (BE), Natura 2000 sites (NL), or mentioned RSCs, such as the Barcelona Convention (IT), OSPAR and HELCOM (DK).

In BAL, PL made references to other descriptors about the reduction of pressures D6C1, D6C2, D6C3 and D7C1 and D7C2. While FI addressed oil and chemical accidents.

In BLK, RO provided a spatial extent for habitat type loss for D6C4 and targets value with indicators for D6C5.

MS did not address all the GES components with the reported targets. For instance, EE uses only at Descriptor, D6C1 and D6C2 levels. FI provided targets only for D6C5, being also the only criteria assessed under Art. 8.

4.3 Adequacy of pressures by regions and subregions

Figure 14 presents the reported number of pressures per pressure group and the proportion of each group addressed by the overall targets by region. The pressures groups are: biological pressures (PresBio), pressure impacting the marine environment (PresEnv), pressures related to inputs of substances, litter and energy (PresInput) and physical pressures (PresPhy).

Data: Member States 2018 reports (file in Annex: D6-Art.10; columns: Features).

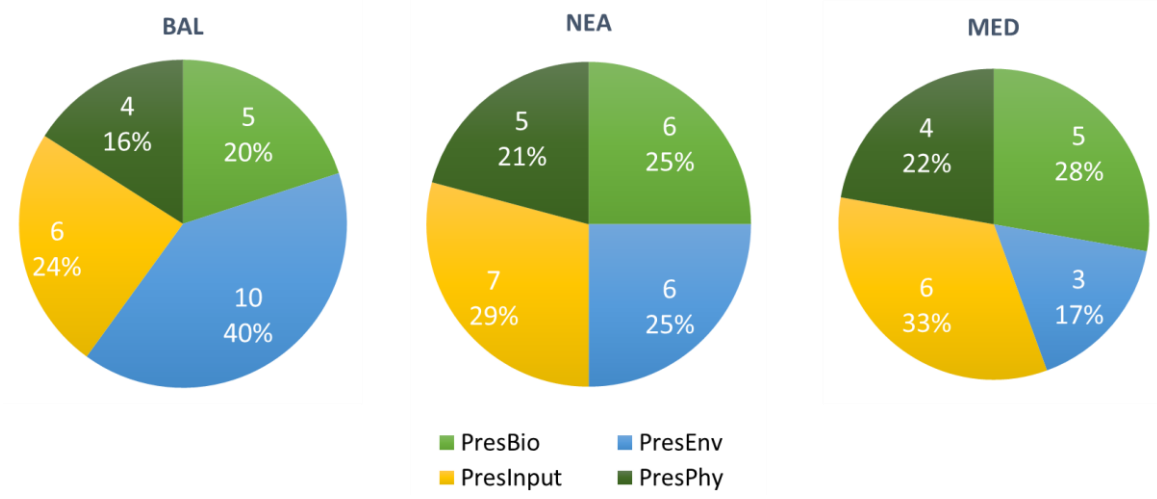

Figure 14. Number of pressures reported for each pressure group and proportion of each pressure group addressed by the overall targets by region in D6.

Findings:

Overall, biological pressures were included in 124 targets (19.2% of overall targets), physical pressures were included in 336 targets (49.9%) reported by 13 MS, input pressures were included in 67 targets (9.4%) reported by five MS, and the environmental pressures were included in 44 targets (6.2%) reported by five MS. Two MS (ES, PL) addressed 'all pressures' with 14 targets (<2.0%).

For the GES component D6C1, seven MS (HR, IE, IT, NL, PL, SE) addressed the 'Physical loss of the seabed' (PresPhyLoss), and three MS (IT, NL, SE) addressed the 'Physical disturbance to seabed' (PresPhyDisturbSeabed)[4].

For D6C2, six MS (HR, IE, IT, NL, PL, SE) addressed the 'Physical disturbance to seabed', three MS (IT, NL, SE) addressed the 'Physical loss of the seabed' and two MS (IT, NL) addressed the 'Extraction of, or mortality/injury to, wild species (by commercial and recreational fishing and other activities)' (PresBioExtractSpp).

For D6C3, four MS (BE, NL, PL, SE) addressed the 'Physical disturbance to seabed'. The 'Physical loss of the seabed' (was addressed by NL and SE, the 'Extraction of, or mortality/injury to, wild species (by commercial and recreational fishing and other activities)' was addressed by NL, and 'All physical pressures' (PresPhyAll) by PL.

For D6C4, NL, PL and SE addressed the pressure 'Physical loss of the seabed' (, NL and SE addressed the pressure 'Physical disturbance to seabed', and NL only addressed the

[4] for the default values of pressures, see EU, 2019.

‘Extraction of, or mortality/injury to, wild species (by commercial and recreational fishing and other activities)’.

For D6C5, BE, NL and SE addressed the ‘Physical disturbance to seabed’, NL and SE addressed the ‘Extraction of, or mortality/injury to, wild species (by commercial and recreational fishing and other activities)’, and FI and PL addressed ‘Input of other substances (e.g., synthetic substances, non-synthetic substances, radionuclides) – diffuse sources, point sources, atmospheric deposition, acute events’ (PresInputCont). PL also addressed ‘Changes to hydrological conditions’ (PresPhyHydroCond), ‘Input of nutrients – diffuse sources, point sources, atmospheric deposition’ (PresInputNut), and ‘Input of organic matter – diffuse sources and point sources‘ (PresInputOrg), while SE addressed ‘Contaminants - UPBT substances’ (PresEnvContUPBTs), ‘Contaminants – in seafood’ (PresEnvContSeafood), ‘Contaminants - non UPBT substances’ (PresEnvContNonUPBTs), ‘Established non-indigenous species’ (PresEnvNISestablished), ‘Newly-introduced non-indigenous species’ (PresEnvNISnew), ‘Litter and micro-litter in species’ (PresEnvLitterSpp), ‘Litter in the environment’ (PresEnvLitter), ‘Physical disturbance to seabed’PresPhyDisturbSeabed’and ‘Changes to hydrological conditions’.

Other pressures were addressed under the GES component D6, i.e., at descriptor level.

In BAL, of the total eight MS, four MS (50%; DE, LT, PL, SE) addressed at least one of the four physical pressures identified, four MS (38%) addressed input (DE, FI, PL), environmental (DE, LT, SE) and biological pressures (DE, FI, SE).

In NEA, of the total nine MS, eight MS (89%), all but DK, addressed physical pressure with five identified pressures, six MS (67%; DE, ES, FR, NL, PT, SE) addressed biological pressures, especially all the MS addressed for ‘Extraction of, or mortality/injury to, wild species (by commercial and recreational fishing and other activities)’; three MS (33%; DE, ES, FR) addressed input pressures and other three MS (33%; DE, FR, SE) addressed environmental pressures.

In MED, of the seven total MS, four MS (57%; CY, ES, HR, IT, FR) addressed physical pressures, three MS (43%; ES, IT, FR) addressed biological pressures, two MS (27%; ES, FR) environmental and one MS (FR) addressed input pressures.

In the BLK, no pressure was reported.

4.4 Analysis of the levels of details of reported targets

Data: Member States 2018 reports (file in Annex: D6-Art.10; columns: element, parameter, other parameter, target value).

4.4.1 Elements

Overall, 132 elements were reported, five MS in BAL reported 39 elements, eight MS in NEA reported 97 elements (mostly the NL), four MS in MED reported 11 elements and one MS in BLK reported six elements.

LV reported as elements only species groups and habitats under the GES component at D6/D1 level.

4.4.2 Parameters

Table 32. Parameters reported by the MS displayed by region.

Parameter	Regions			
	BAL	NEA	MED	BLK
Abundance (ABU)	LV	BE FR NL		
Biomass (BIOM)		FR NL	IT	
Concentration in water (CONC-W)	DE	DE FR	FR	
Distribution (range) (DIST-R)			HR	
Distribution (spatial) (DIST-S)	DE	DE HR		
Extent (EXT)	DE FI LT PL SE	BE DE ES FR IE NL PT SE	ES FR HR IT	RO
Fecundity (breeding rate) (FEC)		NL		
Habitat condition (HAB-CON)	DE DK LV PL	DE DK ES FR IE NL PT	CY ES FR HR IT	RO
Mass (MASS)			FR	
Presence (PRE)	SE	SE		
Size distribution (SIZE-D)	SE	NL SE		

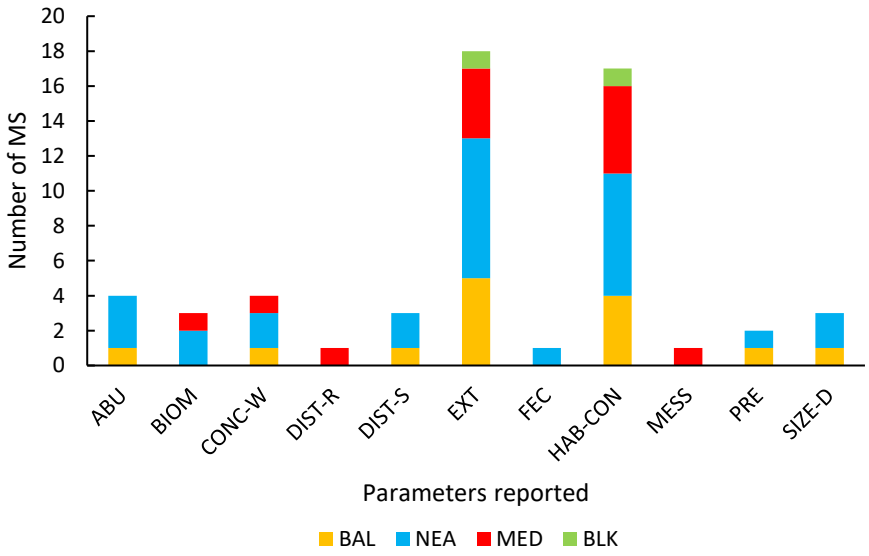

Figure 15. Number of MS reporting the default parameters in D6 Art. 10 by region.

Findings: 594 targets reported a parameter, of which 353 targets (59.4%) reported one of the 11 default parameters present in the 'Parameter' field and 241 targets (40.6%) reported one of the 70 additional parameters reported in the in the 'Parameter other' field.

4.4.3 Target values

In Table 33 the target values are reported per parameter and the number of targets that use the target value is quantified in the last column. The elements are pooled together by target value, region and MS. The target value source is reported under Description target assessment. Top table includes the reporting parameters defined in the reporting guidance (EU, 2019), while bottom table includes the parameters reported as "other parameters".

Table 33. Target values per parameter, with their units, source, group of elements, region and Member State. Top table includes the reporting parameters from the reporting guidance (EU, 2019); bottom table includes the parameters reported as "other parameters".

Parameter	Target value	Value unit	Target value source	Element	Region	MS	No. targets
ABU	0.6			Coastal fish; Demersal shelf fish; Pelagic shelf fish; Seals	BAL	LV	4
ABU	0.0 Trend indicator			Seal species	NEA	NL	6
ABU	75.0		OSPAR	Benthic feeders breeding; Benthic feeders non breeding; Grazing feeders non breeding	NEA	NL	18
CONC-W	2.6	mg/L	WFD	TN	BAL	DE	1
CONC-W	2.8	mg/L	WFD	TN	NEA	DE	1
DIST-S	0.0 Trend indicator			Seal species	NEA	NL	6
EXT	1.0	%		None*	BAL	LT	3
EXT	10.0	%		None*	BAL	LT	1
EXT	0.0	events/y		Habitats	NEA	FR	3
EXT	0.0	%		Area physically lost, disturbed or hydrographically	NEA MED	FR	4
EXT	5.0	%		Circalittoral and Infralittoral habitats; offshore	BLK	RO	8
FEC	-1.0	%		Halichoerus grypus	NEA	NL	2
HAB-CON	0.0	Trend indicator	OSPAR	Used for OBT	NEA	NL	30
HAB-CON	1.0	Trend indicator		Used for BHT	NEA	NL	49
HAB-CON	3.0	%		Habitat	NEA	LV	1
HAB-CON	0.61	M-AMBI(n)		Circalittoral mud (*Modiolula phaseolina*)	BLK	RO	3

Parameter	Target value	Value unit	Target value source	Element	Region	MS	No. targets
HAB-CON	0.68	M-AMBI(n)		Biogenic reefs on circalittoral mud (*Mytillus galoprovincialis*); Circalittoral sand; Infralittoral sand	BLK	RO	4
SIZE-D	1.0	Trend		Demersal fish community	NEA	NL	1

* Parameters related to C1 and C2 des not report element

Other parameter	Target value	Value unit	Element	Region	MS	No. targets
Nitrogen inputs	7670.0	Tonnes per year	TN	BAL	DE	1
Phosphorus inputs	170.0	Tonnes per year	N° OF TP	BAL	DE	1
Distribution	0.6	ratio	Seals	BAL	LV	1
Reproductive and nutritional status	0.6	ratio	Seals	BAL	LV	1
Number	0.0	events/y %	Habitats	NEA	FR	9
Ratio	0.0	%	Habitats	NEA	FR	2
Breeding Success	3.0	Broods per 6 years	Bird species	NEA	NL	38
Proportion failing to spawn	1.0	Trend indicator	Sensitive fish species	NEA	NL	1
Number	0.0	%	Species	MED	FR	1
Number	100.0	%	Habitat types	MED	FR	1
Number	2.0	Number	Habitat types	MED	FR	1
Number	3.0	Number	Habitat types	MED	FR	1

Findings:

70% of the reported parameters in the 'Parameter' field have one or more target values and some have a value source specified in the field 'Description target assessment'. Some of the targets are not relevant for D6 (e.g., targets related to seals, fish, birds).

Gaps: 16.8% (120) of targets had no methodology (element, parameter, target value) reported to assess target achievement.

4.5 General recommendations for Art. 10

Gaps: The analysis revealed data gaps, which are evident in the difficulty of the MS to provide measurable targets focused on particular features and elements. However, it is more urgent to develop guidelines on how the MS should develop and set targets in a consistent way, relevant to the GES determination, measurable (able to measure the distance between current status and GES) and linked with the measures reported under Art. 13. The reported elements do not follow a consistent pattern across the MS and similarly with the fluctuating number of targets reported per MS the elements are inconsistently assigned across MS. Some target values are set, without a regional coordination on the target value and the necessary response (measures); currently, targets do not focus on the relevant pressures at national/regional scale and do not allow comparison across MS. All target values correspond to indicators or criteria that were reported for Art. 8 and do not refer to measurable targets able to estimate the gap between current state and GES.

Recommendation for reporting: The reporting structure provides all the required information to evaluate targets, as such there are no recommendations for refining the reporting, especially when consistency and common understanding on Art. 10 requirements remain unsolved.

Recommendation for implementation: A guidance document focused on MSFD Art. 10 target is required, following a similar organisation to the Art. 8 guidance, to ensure a common understanding across the MS and enforce regional coordination for setting regional targets. This would greatly facilitate the monitoring and evaluation of the GES achievement for D6 at the proper ecological scale and at regional or subdivisional level.

The SWD(2020)62[5] (European Commission, 2020) provides concrete recommendation and guidance for Art. 10 implementation and reporting, built on the experience gained during the first MSFD cycle and with the contribution of the MS. This document can facilitate a common understanding across the MS; however, our evaluation of the MS reports for D6 showed that it was not consistently used by the MS.

In any case, a consistent and accurate GES determination for Art. 9 can streamline the consistency and harmonisation of Art. 10, as well.

[5] This document was published in 2020 (after 2018 reporting deadline) however it was widely available to MS as the CCI paper with various versions released to WG GES and MSCG during 2015-2017. It is expected that the document will be used in the next cycle of MSFD reporting.

5 Conclusions

Generally, the MS put a huge effort to fulfil the reporting obligation for the MSFD updates for Art. 8, 9 and 10 for D6/D1 – benthic habitats. The vast amount of the reported information was very challenging to be grouped and analysed indicating that there is room for improving the reporting process and guidance. Moreover, the consistency and quality of the reported information can be improved with regional cooperation, harmonisation of the methods and a common understanding on the key concepts of the implementation for benthic habitat assessments, GES determination and target setting.

- For Art. 8, few threshold values were reported despite the expectation that several benthic habitats were assessed under other policies (HD, WFD) or the RSCs. Similarly, very few completed assessments, based on an assessed value and on an estimated value were made for some criteria and BHT. Many gaps exist in the assessment methodology across BHT in all regions. A regional or EU agreement on the assessment methods by ecologically relevant areas (subdivision), threshold values, integration rules and common assessment approach will greatly harmonise and provide comparable assessments. The TG Seabed can support this work. The progress made by the TG Seabed could not be reflected in the evaluated reporting cycle as most of the MS completed their assessments before the establishment of the TG, however, many of the above issues are already discussed and promising recommendations for comparable assessments can inform the updates of Art. 8 guidance.
- For Art. 9, many different and inconsistent GES determinations were observed. The inconsistency was caused by the lack of a common approach to determine GES and lack of data to support the GES determination. The GES determination should be harmonized at regional level in order to provide a comparable assessment for benthic habitats. The assessment flow and the Art. 8 guidance should be the basis for building a common GES determination in a consistent way and agreeing how the GES will be reflected at the level of features, indicators and criteria. Since the MS invested considerable effort to harmonise methods for assessments, that improved the common understanding of the MSFD requirements, the next step would be to build on this common understanding and harmonise GES determination. Common integration rules and common agreed methods will bring consistency in all levels. The MSFD reporting tool (EEA's Reportnet 3.0) should be developed accordingly, linking GES determination to the proper level reducing and automatizing the reporting effort. Starting from the technical groups, the MS' appointed experts could propose how to formulate a common approach for a quantitative GES determination at regional or subregional level, which then could be endorsed by the MS through the MSFD CIS groups. A harmonised GES determination can guide the reporting process on the required information to be included (e.g., elements to assess, parameter used and threshold values, integration type).
- For Art. 10, the analysis revealed data gaps, which are evident in the difficulty of the MS to provide measurable targets for all criteria. Targets do not focus on relevant pressures at national/regional level and there is no regional coordination on the targets setting and response (measure). There is a complete lack of measurable targets with associated target values for the reported parameters. Thus, the targets are unable to measure the gap between the current state and GES. A guidance document focused

on MSFD Art. 10 target is required, following a similar organisation to the Art. 8 guidance, to ensure a common understanding across the MS and enforce a regional coordination for setting regional targets. This would greatly facilitate the monitoring and evaluation of the GES achievement for D6/D1 – benthic habitats, at the proper ecological scale and at regional or subdivisional level.

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List of abbreviations and definitions

ABI	Atlantic - Bay of Biscay and Iberian Coast
ACS	Atlantic - Celtic Seas
AMA	Atlantic - Macaronesia
ANS	Atlantic - Greater North Sea
BAL	Baltic Sea
BARCON	Barcelona Convention for the Protection of the Marine Environment and the Coastal Region of the Mediterranean
BD	Birds Directive
BLK	Black Sea
EC	European Commission
EU	European Union
EUNIS	European Nature Information System
GES	Good Environmental Status
HELCOM	Convention on the Protection of the Marine Environment of the Baltic Sea Area
ICES	International Council for the Exploration of the Sea
MAD	Mediterranean - Adriatic Sea
MIC	Mediterranean - Ionian Sea and Central Mediterranean Sea
MRU	Marine Reporting Unit
MS	Member State
MSFD	Marine Strategy Framework Directive
MWE	Mediterranean - Western Mediterranean Sea
NIS	Non-indigenous species
OSPAR	Convention for the Protection of the Marine Environment of the Northeast Atlantic
RSC	Regional Sea Convention
UNEP/MAP	United Nations Environment Programme - Mediterranean Action Plan
WFD	Water Framework Directive

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Annex I

The spreadsheets containing the reported information on which the evaluation was based are embedded below. Each spreadsheet (or group of spreadsheets) corresponds to Art. 8, 9 and 10 respectively.

- Art. 8 reported data

D6-Art.8.xlsx

- Art. 9 reported data

D6-Art.9.xlsx

- Art. 10 reported data

D6-Art.10.xlsx

Annex II

An update on the new EUNIS marine classification, on future developments in EMODnet to provide full coverage seabed habitat maps for Europe, according to the EUNIS classification and MSFD broad habitat types, and an update on the distribution of MSFD broad habitat types per region, subregion and Member State. This annex is included in a document prepared by the TG Seabed (Circabc[6], SEABED_3-2020-05).

Region	Baltic Sea								North-east Atlancic Ocean														Mediterranean Sea													Black Sea	
Subregion									Greater North Sea							Celtic Seas			Bay of Biscay & Iberian Coast			Macaronesia	Western Mediterranean Sea				Ionian Sea & Central Mediterranean Sea			Adriatic Sea				Aegean-Levantine Sea			
MSFD broad habitat type	FI	EE	LV	LT	PL	DE	DK	SE	SE	DK	DE	NL	BE	FR	UK	UK	IE	FR	FR	ES	PT	PT	ES	ES	UK	FR	IT	IT	MT	EL	IT	SI	HR	EL	EL	CY	BG	RO
Littoral rock and biogenic reef											Y			Y			Y	Y	Y		Y	Y				Y						Y	Y				Y	Y
Littoral sediment											Y	Y		Y			Y	Y	Y		Y	Y				Y						Y	Y				Y	Y
Infralittoral rock and biogenic reef	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y			Y	Y	Y	Y	Y	Y	Y	Y	Y	Y		Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Infralittoral coarse sediment			Y			Y					Y			Y			Y	Y	Y		Y					Y						Y	Y				Y	Y
Infralittoral mixed sediment	Y	Y	Y		Y	Y	Y	Y	Y	Y	N			Y	Y	Y	Y	Y	Y	Y	Y	Y				Y	Y	Y	Y		Y	Y	Y		Y	Y	Y	Y
Infralittoral sand	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Infralittoral mud	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y		Y	Y	Y	Y	Y	Y	Y	Y		Y	Y	Y	Y
Circalittoral rock and biogenic reef	Y	Y	Y	Y	Y	Y	Y	Y	Y		Y			Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y		Y	Y	Y			Y	Y	Y
Circalittoral coarse sediment	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y		Y	Y	Y	Y		Y	Y	Y		Y	Y	Y	Y
Circalittoral mixed sediment	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y				Y	Y	Y	Y		Y	Y	Y		Y	Y	Y	Y
Circalittoral sand	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Circalittoral mud	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y		Y	Y
Offshore circalittoral rock and biogenic reef			Y						Y		Y			Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y			Y		Y			Y	N	N
Offshore circalittoral coarse sediment	Y	Y	Y		Y		Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y		Y	Y	Y			Y				Y		Y	N
Offshore circalittoral mixed sediment	Y	Y	Y	Y	Y	N	Y	Y	Y	Y	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y					Y	Y	Y		Y				Y		Y	Y
Offshore circalittoral sand	Y	Y	Y	Y	Y	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y		Y	Y		Y	Y	Y	Y	Y	Y
Offshore circalittoral mud	Y	Y	Y	Y	Y	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	N	Y	Y	Y		Y	Y
Upper bathyal rock and biogenic reef									Y							Y	Y	Y	Y	Y	Y	Y	Y	Y		Y	Y	Y	Y		Y					Y	N	Y
Upper bathyal sediment									Y	Y						Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y		Y	Y	Y	Y	Y	Y
Lower bathyal rock and biogenic reef																Y	Y	Y	Y	Y	Y	Y	Y	Y		Y	Y	Y	Y		Y					Y	N	?
Lower bathyal sediment																Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y				Y	Y	Y	Y
Abyssal																Y	Y	Y	Y	Y	Y	Y	Y	Y		Y	Y	Y	Y	Y					Y		Y	Y
Checked			Ingrida Purina			Axel Kreutle					Axel Kreutle	Sander Wijnhoven		Hélène Muller			Oliver Ó Cadhla	Hélène Muller	Hélène Muller		Jorge Lobo Arteaga	Cláudia Ribeiro; Andreia Braga Henriques				Hélène Muller	Saša Raichevich	Saša Raichevich		Nadia Papadopoulou	Saša Raichevich	Borut Mavrič	Marija Despalatović	Nadia Papadopoulou	Nadia Papadopoulou		Valentina Todorova	Valeria Abaza
Date			25/09/2019			19/11/2019					19/11/2019	25/09/2019		02/09/2019			17/06/2020	02/09/2019	02/09/2019		28/10/2019	05/11/2019				02/09/2019	16/07/2019	16/07/2019		03/09/2019	16/07/2019	13/06/2019	25/10/2019	03/09/2019	03/09/2019		28/06/2019	05/07/2019

Key						
Habitat present (Y) or absent (blank) based on 2016 EMODnet MSFD broad habitat	Y	Y	Y	Y	Y	
Habitat present (Y) or absent (N) based on feedback from TG Seabed members	Y	Y	Y	Y/N	Y	N

[6] https://circabc.europa.eu/w/browse/85f995b6-819a-4d1b-aecd-630df32f83d6

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KJ-NA-30716-EN-N

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doi:10.2760/355956

ISBN 978-92-76-38014-6